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Semaphorin7A patterns neural circuitry in the lateral line of the zebrafish

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eLife assessment

Dasgupta and colleagues make a **valuable** contribution to the understanding how the guidance factor Sema7a promotes connections between mechanosensory hair cells and afferent neurons of the zebrafish lateral line system. The authors provide **solid** evidence that loss of Sema7a function results in fewer contacts between hair cells and afferents through comprehensive quantitative analysis. Additional work is needed to distinguish the effects of different isoforms of Sema7a to determine whether there are specific roles of secreted and membrane bound forms.

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Abstract

In a developing nervous system, axonal arbors often undergo complex rearrangements before neural circuits attain their final innervation topology. In the lateral line sensory system of the zebrafish, developing sensory axons reorganize their terminal arborization patterns to establish precise neural microcircuits around the mechanosensory hair cells. However, a quantitative understanding of the changes in the sensory arbor morphology and the regulators behind the microcircuit assembly remain enigmatic. Here, we report that Semaphorin7A (Sema7A) acts as an important mediator of these processes. Utilizing a semi-automated three-dimensional neurite tracing methodology and computational techniques, we have identified and quantitatively analyzed distinct topological features that shape the network in wild-type and Sema7A loss-of-function mutants. In contrast to those of wild-type animals, the sensory axons in Sema7A mutants display aberrant arborizations with disorganized network topology and diminished contacts to hair cells. Moreover, ectopic expression of a secreted form of Sema7A by non-hair cells induces chemotropic guidance of sensory axons. Our findings propose that Sema7A likely functions both as a juxtracrine and as a secreted cue to pattern neural circuitry during sensory organ development.

Introduction

Pathfinding axons are directed to their appropriate synaptic targets by a variety of guidance cues. While approaching or traversing the target field, the growth cones of migrating axons encounter secreted or cell surface-attached ligands and respond by steering toward or away from their sources (1–4). Fine-grained control of these factors is critical for the establishment of proper neural circuitry.

The zebrafish's lateral line provides a tractable *in vivo* system for exploring the mechanisms that guide the assembly of neural circuits in a peripheral nervous system. In particular, it is possible to detect individual hair cells and the sensory axons that innervate them throughout the first week of development (5). The primary posterior lateral line on each side of the zebrafish's tail consists of about seven neuromasts, which contain mechanoreceptive hair cells of opposing polarities—half sensitive to headward (rostrad) water motion and the complementary half to tailward (caudad) water motion—that cluster at their centers (6). The sensory axons of the lateral-line nerve branch, arborize, and consolidate around the basolateral surfaces of the hair cells (**Figure 1A**). To aid in an animal's swimming behavior, these hair cells sense water currents and relay signals to the brain through the sensory axons (7). The well-defined structure of the ramifications and synaptic contacts of the sensory axons highlight the need to determine the molecular signals behind the assembly of such precise microcircuits (8).

We adopted a candidate-gene strategy to seek factors that direct the growth of afferent growth cones in the zebrafish's lateral line. Analysis of single-cell RNA sequencing data identified the *semaphorin7a* (*sema7a*) gene to be highly expressed in hair cells of that organ (9). Semaphorins are important regulators of axonal growth and target finding during the patterning of diverse neural circuits (10). The semaphorin family includes proteins that are secreted, transmembrane, and cell surface-attached (11). Among these, Semaphorin7A (Sema7A) is the only molecule that is anchored to the outer leaflet of the lipid bilayer of the cell membrane by a glycosylphosphatidylinositol (GPI) linker (11). Unlike many other semaphorins, which act as repulsive cues (12,13), Sema7A is involved in promoting axon growth (4) and imparting directional signals by interacting with the integrin and plexin families of receptors residing on the pathfinding axons (14–16). Sema7A likely occurs *in vivo* in both GPI-anchored and soluble forms (17–19), and studies of neuronal explants confirm that both forms can induce directed axonal outgrowth (4). Furthermore, it has been proposed that the GPI anchor is cleaved by membrane-resident GPI-specific phospholipases (GPI-PLs) or matrix metalloproteases to release the Sema7A into the extracellular environment and thus regulate dynamic cellular processes (20,21). Because these observations indicate that Sema7A might influence neuronal development both as a juxtracrine and as a diffusive signal, we investigated the role of Sema7A in sculpting a vertebrate peripheral sensory organ.

Results

Sema7A expression and localization in hair cells

The zebrafish genome contains a single *sema7a* gene that produces two transcripts. One transcript encodes Sema7A-GPI, a full-length, GPI-linked, cell surface-attached protein. The second transcript yields Sema7A^{sec}, a protein with a truncated C-terminus, which we conjecture is secreted into the local environment. Each transcript encodes an N-terminal signal sequence and a single copy of the conserved Sema domain (11). To detect the presence of both *sema7a* transcripts in the hair cells of the lateral-line neuromasts, we used fluorescence-activated cell sorting (FACS) to capture the

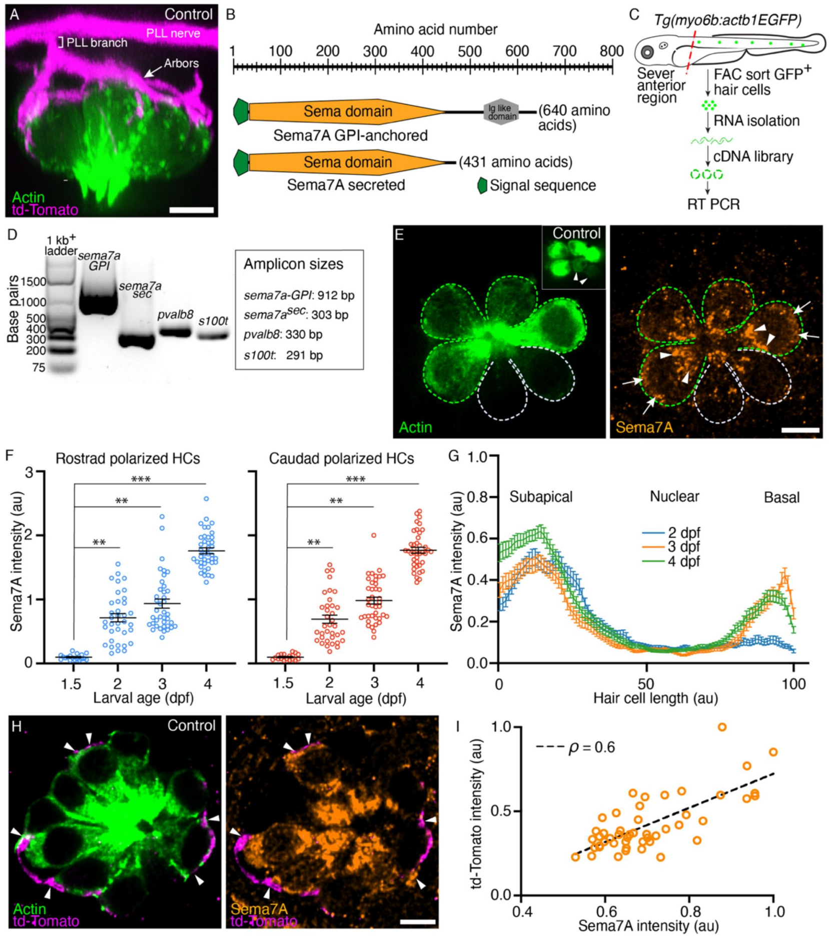

Figure 1.

Expression of Semaphorin7A in the zebrafish's lateral line.

(A) A volumetric rendering of a PLL neuromast depicts the sensory axons (magenta) that branch from the lateral line nerve to arborize around the basolateral surface of the hair cell cluster (green). Additional cell types in the neuromast are not labeled. Larval age, 3 dpf. (B) A schematic drawing of the two variants of the Sema7A protein molecule depicts the full-length GPI-anchored form and the smaller, potentially secreted form. Both the molecules include a signal sequence (green) and a conserved sema domain (orange). (C) *Tg(myo6b:actb1-EGFP)* larvae at 4 dpf were terminally anesthetized and their heads were removed with fine blades. The tails were then dissociated into the component cells, which were sorted by flow cytometry. Green cells represent GFP$^+$ hair cells. A cDNA library was prepared from the isolated RNA. (D) Gel-based RT-PCR analysis indicates the presence of both the *sema7a* transcript variants and the expression of hair cell-specific genes, parvalbumin 8 (*pvalb8*) and S100 calcium binding protein T (*s100t*). (E) A surface micrograph of a neuromast at 3 dpf depicts two pairs of mature hair cells (green dashed lines) and a pair of immature hair cells (grey dashed lines). Inset: among the three pairs of hair cell apices, the immature pair is indicated by arrowheads. Immunolabeling reveals that the Sema7A protein (orange) occurs consistently at the subapical region (arrowheads) and at the basolateral surface (arrows) of a hair cell. In this and in each of the subsequent neuromast images, anterior is to the left and dorsal to the top. (F) A plot quantifies developmental changes in the average Sema7A intensity in both rostrally and caudally polarized hair cells of neuromasts from 1.5 dpf to 4 dpf. The data stem from 18, 36, 39, and 40 hair cells in neuromasts of respectively 1.5 dpf, 2 dpf, 3 dpf, and 4 dpf larvae. (G) A plot quantifies the distribution of average Sema7A intensity along the hair cell's apicobasal axis. The results stem from 52, 57, and 54 hair cells of neuromasts from 2 dpf, 3 dpf, and 4 dpf larvae. (H) An immunofluorescence image at the nuclear level of a 4 dpf neuromast shows the contact of the sensory arbors (magenta) with the basolateral surface of the hair cells (green). Immunolabeling for Sema7A (orange) reveals enrichment of the protein at the hair cell bases and sensory-axon interfaces (arrowheads). (I) At the base of the hair cell, association of the td-Tomato$^+$ sensory arbors positively correlated with the Sema7A intensity. The results stemmed from 48 hair cells of 13 neuromasts from 3 dpf larvae. HC, hair cell; scale bars, 5 µm; au, arbitrary unit; means ± SEMs; ρ, Spearman's correlation coefficient; *** implies $p < 0.001$ and ** implies $p < 0.01$.

labeled hair cells (22) and isolated total RNA. Using primers that flank the regions encoding the conserved Sema domain and the distinct C-termini of the two transcripts, we performed reverse transcription and polymerase chain reactions (RT-PCR) to identify both the membrane-anchored and the secreted transcript in developing larvae (**Figure 1B-D**; **Figure 1—figure supplement 1A-D**). Single-cell RNA sequencing data also have shown that the GPI-linked *sema7a* transcript is particularly enriched in the immature and mature hair cells of the neuromast during early larval development (9), a period when sensory axons of the posterior lateral line contact hair cells and form synapses with them (5,23). Indeed, we have observed Sema7A to be specifically localized in both mature and immature hair cells (**Figure 1E**; **Figure 1—figure supplement 2A-D**).

As neuromasts mature, the contacts between sensory axons and hair cells increase in number, become stabilized, and establish synapses (5,23). Because Sema7A can modulate axon guidance (4,14) and synapse formation (15), we wondered whether the level of Sema7A also changes during neuromast development. Upon quantifying the average intensity of Sema7A, we found similar expression in rostrally and caudally polarized hair cells of neuromasts at 1.5, 2, 3, and 4 days post fertilization (dpf). The average Sema7A intensity increased significantly over this period in both rostrally and caudally polarized hair cells. At 1.5 dpf, the neuromasts harbor primarily immature hair cells that make minimal contacts with sensory axons and do not form stable association with axonal terminals (23). At this stage the average Sema7A intensity in hair cells remained low. By 2-4 dpf the hair cells mature to form stable contacts and well-defined synapses with the sensory axons (5,24). The average Sema7A intensity in hair cells at each of these stages rose to levels that significantly exceeded those of 1.5 dpf neuromasts (**Figure 1F**). This rise in the amount of Sema7A during neuromast maturation supports a role for Sema7A in the guidance of sensory axons and their interaction with hair cells.

We additionally demonstrated an anisotropic distribution of Sema7A along the apicobasal axis of each hair cell. As neuromasts develop through 2, 3, and 4 dpf, Sema7A remained highly enriched in the subapical region of the hair cell, the site of the Golgi network (**Figure 1E**). This localization suggests that Sema7A—like other proteins of the semaphorin family (25)—is directed to the plasmalemma through Golgi-mediated vesicular trafficking. Using the transgenic line *Tg(cldnb:LY-mScarlet)* that expresses membrane-tethered mScarlet in all cells of the neuromast (26), we detected Sema7A at the plasma membrane and potentially in vesicles (**Figure 1—figure supplement 2E-H**). While Sema7A maintained a high level in the subapical region, the protein increasingly accumulated at the base of the hair cell: the Sema7A level at the hair cell base was low at 2 dpf, but increased sharply by 3 dpf and 4 dpf (**Figure 1G**; **Figure 1—figure supplement 1I**,**J**). During this period the hair cells increase in number and mature, whereas the associated sensory axons arborize to contact multiple hair cells and form robust synaptic boutons at their bases (5,24). Does progressive basal accumulation of Sema7A facilitate association of sensory axon terminals to the hair cell? To address this question, we utilized doubly labeled transgenic fish *Tg(myo6b:actb1-EGFP;neurod1:tdTomato)* that marked the hair cells and the sensory axons of the posterior lateral line with distinct fluorophores (**Figure 1A**) (27). We quantified the average Sema7A intensity and the accumulation of sensory arbors, measured by td-Tomato intensity, at the bases of the hair cells. Across developmental stages, the basal association of the sensory arbors showed a positive correlation with Sema7A accumulation (**Figure 1H**,**I**; **Figure 1—figure supplement 3A-E**, Video 1). The progressive enrichment of Sema7A at the hair cell base with the sensory axons suggests that Sema7A acts as a mediator of contacts between sensory axons and hair cells.

Sema7A restricts sensory axons around clustered hair cells

If Sema7A guides and restricts the sensory arbors at and around the hair cells clustered in a neuromast, then inactivating Sema7A function should disrupt this process. To test this hypothesis, we obtained a *sema7a* mutant allele (*sema7a*^{sa24691}) with a point mutation that introduces a premature stop codon in the conserved sema domain (**Figure 2—figure supplement 1A**). The homozygous mutant, hereafter designated *sema7a*$^{-/-}$, should lack both the secreted and the membrane-attached forms of Sema7A. The average intensity of Sema7A immunolabeling in hair cells diminished by 61% in *sema7a*$^{-/-}$larvae (0.34 ± 0.01 arbitrary units) in comparison to controls (0.87 ± 0.01 arbitrary units). In a few cases we observed minute accumulations of Sema7A at the subapical region, but little or no protein at the base of the mutant hair cells (**Figure 2A**; **Figure 2—figure supplement 1B**). Furthermore, we tested the mechanotransduction ability of *sema7a*$^{-/-}$ mutant hair cells by labeling with the styryl dye FM4-64 (5). The *sema7a*$^{-/-}$ larvae incorporated the FM4-64 as effectively as the control animals. This result suggests that the mutation does not impact essential hair cell functions (**Figure 2—figure supplement 1C**,**D**, Videos 2,3).

To characterize the impact of the *sema7a*$^{-/-}$ mutation on arbor patterning, we utilized doubly labeled transgenic fish *Tg(myo6b:actb1-EGFP;neurod1:tdTomato)* that distinctly labeled the hair cells and the sensory arbors (27). In control neuromasts, the sensory axons approached, arborized, and consolidated around the bases of clustered hair cells (**Figure 2B**). Although the sensory axons approached the hair cells of *sema7a*$^{-/-}$ neuromasts, they displayed aberrant arborization patterns with wayward projections that extended transversely to the organ (**Figure 2C**). To quantitatively analyze the arborization morphology, we traced the sensory arbors in three dimensions to generate skeletonized network traces that depict–as pseudocolored trajectories–the increase in arbor length from the point of arborization (**Figure 2D**, **E**).

To visualize the hair cell clusters and associated arborization networks from multiple neuromasts, we aligned images of hair cell clusters, registered them at their hair bundles, from neuromasts at 2 dpf, 3 dpf, and 4 dpf (**Figure 2—figure supplement 1E**,**F**). Throughout development, the arborization networks of control neuromasts largely remained within the contours of the hair cell clusters; the few that reached farther nonetheless lingered nearby (**Figure 2F**; **Figure 2—figure supplement 1G**,**I**). This result suggests that an attractive cue retained the axons near the sensory organ. In *sema7a*$^{-/-}$neuromasts, however the neuronal arbors failed to consolidate within the boundaries of hair cell clusters and extended far beyond them (**Figure 2G**; **Figure 2—figure supplement 1H**,**J**; Video 4).

We quantified the distribution of the sensory arbors around the center of the combined hair cell clusters. For both the control and the *sema7a*$^{-/-}$ neuromasts, the arbor densities peaked proximal to the boundaries of the hair cell clusters. Beyond the cluster boundaries, in control neuromasts at 2 dpf, 3 dpf, and 4 dpf the arbor densities fell sharply at respectively 31.7 μm, 36.6 μm, and 32.7 μm from the center. In *sema7a*$^{-/-}$ neuromasts of the same ages, the sensory arbors extended respectively 38.5 μm, 51.4 μm, and 56.6 μm from the center (**Figure 2H**; **Figure 2—figure supplement 1K**,**L**; **Figure 2—figure supplement 2A**). Furthermore, we quantified the degree of contact of the sensory arbors to their hair cell clusters in individual neuromasts from both control and *sema7a*$^{-/-}$mutants. In control neuromasts, the degree of contact was 83 ± 1 % (mean ± SEM) at 2 dpf and 84 ± 1 % at 3 dpf, but significantly increased to 90 ± 1 % at 4 dpf (**Figure 2I**; **Figure 2—figure supplement 2B**). This observation indicates that the sensory arbors reinforce their association to the hair cell clusters as neuromasts develop. However, in *sema7a*$^{-/-}$ neuromasts such contact was significantly reduced. Only 69 ± 4 % of sensory arbors at 2 dpf were closely associated with the corresponding hair cell clusters, a value that remained at 67 ± 2 % at both 3 dpf and 4 dpf (**Figure 2I**). These findings signify that, with simultaneous disruption of both signaling modalities of Sema7A, the organ fails to restrict the localization of sensory arbors and has significantly diminished contact with them.

Sema7A patterns the arborization network's topology

Accurate spatial arrangement of the neuronal projections around hair cells is essential for proper retrieval and transmission of sensory information. Mutations that perturb proper assembly of sensory neuronal microcircuits lead to hearing impairment (28). Quantitative insight into the wiring pattern is therefore essential to identify systematic aberrations in the microcircuit arising

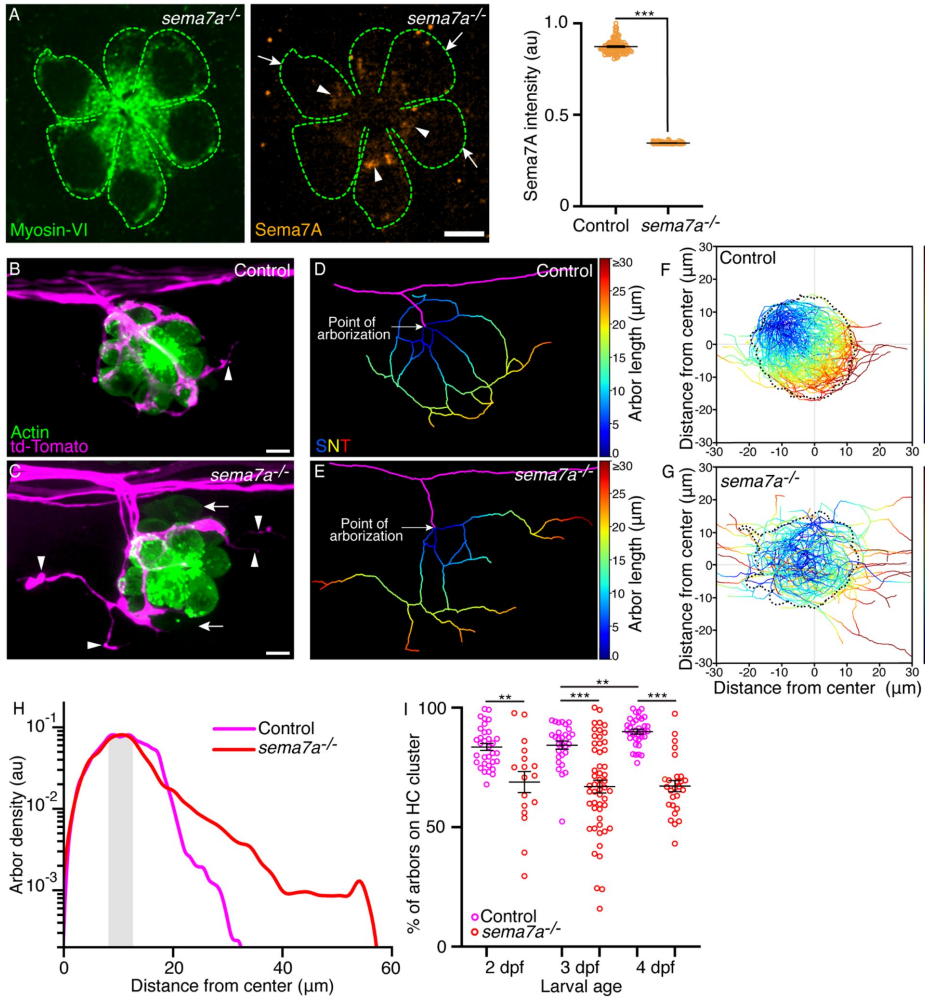

Figure 2.

sema7a$^{-/-}$ mutants display aberrant sensory axon arborizations.

(A) In a micrograph of a 3 dpf *sema7a*$^{-/-}$ neuromast, Myosin VI (green) marks the hair cells (green dashed lines) in which the level of Sema7A (orange) is highly reduced, with sporadic localization in the subapical region (arrowheads) and none in the basolateral (arrow) region. A plot of normalized Sema7A intensity from 99 control and 100 *sema7a*$^{-/-}$ hair cells quantitates the effect. (B,C) Surface views of a control and a *sema7a*$^{-/-}$ neuromast at 4 dpf depict the interaction of the sensory arbors (magenta) with hair cell clusters (green). In the control, the arbors intimately contact the hair cells, with a few exceptions (arrowhead). In the *sema7a*$^{-/-}$ mutant, the arbors direct many aberrant projections (arrowheads) away from the hair cell cluster. The two immature hair cell pairs in the *sema7a*$^{-/-}$ neuromast are indicated by arrows. (D,E) Skeletonized networks portray the three-dimensional topology of the sensory arbors from the control and the *sema7a*$^{-/-}$ neuromasts depicted in panels B and C. The pseudocolored trajectories depict the increase in arbor contour length from each point of arborization, defined as the point at which the lateral line branch (magenta) contacts hair cell cluster. (F,G) Micrographs depict the skeletonized networks of the combined 4 dpf hair cell clusters, whose centers are located at (0,0). The X-and Y-coordinates represent the anteroposterior (AP) and the dorsoventral (DV) axes of the larva, respectively. Positive values of the X-and Y-coordinates represent the posterior and ventral directions, respectively. Combined skeletonized network traces from 27 control and 27 *sema7a*$^{-/-}$ mutant neuromasts are represented. H) The plot denotes the densities of the sensory arbors around the center of the combined hair cell clusters for 35 control (magenta) and 27 *sema7a*$^{-/-}$ (red) neuromasts at 4 dpf. The shaded area marks the region proximal to the boundary of the combined hair cell cluster. (I) The plot quantifies the degree of contact of the sensory arbors to their hair cell clusters in individual neuromasts from both control (magenta) and *sema7a*$^{-/-}$ mutants (red), each point represents a single neuromast. Thirty-three, 29, and 35 neuromasts were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively. Seventeen, 53, and 27 neuromasts were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$ mutant larvae, respectively. HC, hair cell; Scale bars, 5 µm; au, arbitrary unit; means ± SEMs; *** signifies $p < 0.001$; ** signifies $p < 0.01$.

from genetic deficits. The three-dimensional skeletonized network traces allowed us to accurately delineate the microcircuit topology across individuals, developmental stages, and genotypes from living animals (**Figure 2D**, **E**). To quantitatively analyze the arbor network, we defined five topological attributes: (1) nodes, denoted as junctions where arbors branch, converge, or cross; (2) loops, comprising minimal arbor cycles forming topological holes; (3) bare terminals, defined as arbors with free ends; (4) the node degree, calculated as the number of first-neighbor nodes incident to a node; and (5) the arbor curvature, measured as the total curvature of a bare terminal (**Figure 3A**). We quantified these attributes to accurately determine how the circuit topology is established during maturation of the sensory organ in control larvae and how it is impacted by *sema7a* deficit.

To determine how internode connectivity is distributed across networks, we quantified the distribution of node degrees along the relative height of the hair cell cluster across individuals, developmental stages, and genotypes. As for most other real-world networks (29), we observed a highly skewed node-degree distribution: the majority of nodes had low node degrees (three and four) and a small number of nodes—often acting as hubs, like the arborization initiation point—had high node degrees (five to seven). The node degree varied from three to seven in control larvae and three to six in *sema7a*$^{-/-}$ larvae. Across developmental stages and genotypes, the high node degrees primarily occurred at the base (region from 0 % to 10 % of the cluster height) of the hair cell cluster and acted as hubs for the majority of branching events. Low node degrees primarily occurred along the basolateral region (region from 10 % to 60 % of the cluster height) of the hair cell cluster. The apical region (region from 60 % to 100 % of the cluster height) was sparsely populated with low node degrees. The network's hierarchical organization along the organ's axis suggests that arbors receive sensory inputs at multiple levels, allowing the network to carry out robust neural computations. Even though nodes of low degree were most prevalent between genotypes across developmental stages, their distribution along the relative organ axis became significantly different at 4 dpf, with nodes in control larvae localized higher along the axis than in *sema7a*$^{-/-}$ larvae (**Figure 3A**; **Figure 3—figure supplement 1A-C**; **Table 1**). These observations indicate that Sema7A regulates both the spatial organization and node-degree complexity of the sensory arbor network.

An average of 9.3 sensory axonal branches approach the hair cell cluster of a neuromast (8). Pathfinding of sensory axons around the hair cell cluster results in the formation of axonal fascicles that form numerous loops and bare terminals, which determine the microcircuit topology (**Figure 2B**, **D**) (8). To optimally surround and contact the hair cell cluster, do the sensory arbors prefer to adopt one topological feature over the other? To address this question, we measured the correlation between the number of bare terminals and the number of loops across individuals, developmental stages, and genotypes. In both 2 dpf and 3 dpf larvae, the distributions remained highly scattered across individuals and genotypes. However, by 4 dpf in the control larvae the arbor network contained either a high number of loops with a low number of bare terminals or a high number of bare terminals with a low number of loops. This result suggests that the stereotypical circuit topology observed in the mature organ may emerge through transition of individual arbors from forming bare terminals to forming closed loops encircling topological holes. That we did not observe a similar transition in *sema7a*$^{-/-}$ neuromasts of the same age suggests that Sema7A may direct the switch between these two distinct topological features (**Figure 3B**; **Figure 3—figure supplement 1D**, **E**; **Table 1**).

Measurement of the contour length of both bare terminals and loops additionally revealed that across developmental stages in control larvae the sensory arbors followed a three-step process to choose the appropriate topological feature: arbors shorter than 10 μm always formed bare terminals; arbors between 10 μm and 30 μm began to switch between features, with arbors longer than 20 μm strongly preferring to form loops; and arbors longer than 30 μm always formed loops. These results imply that the switch between bare terminals and loops is partly dependent on the length of the arbors. However, this characteristic topological preference was significantly

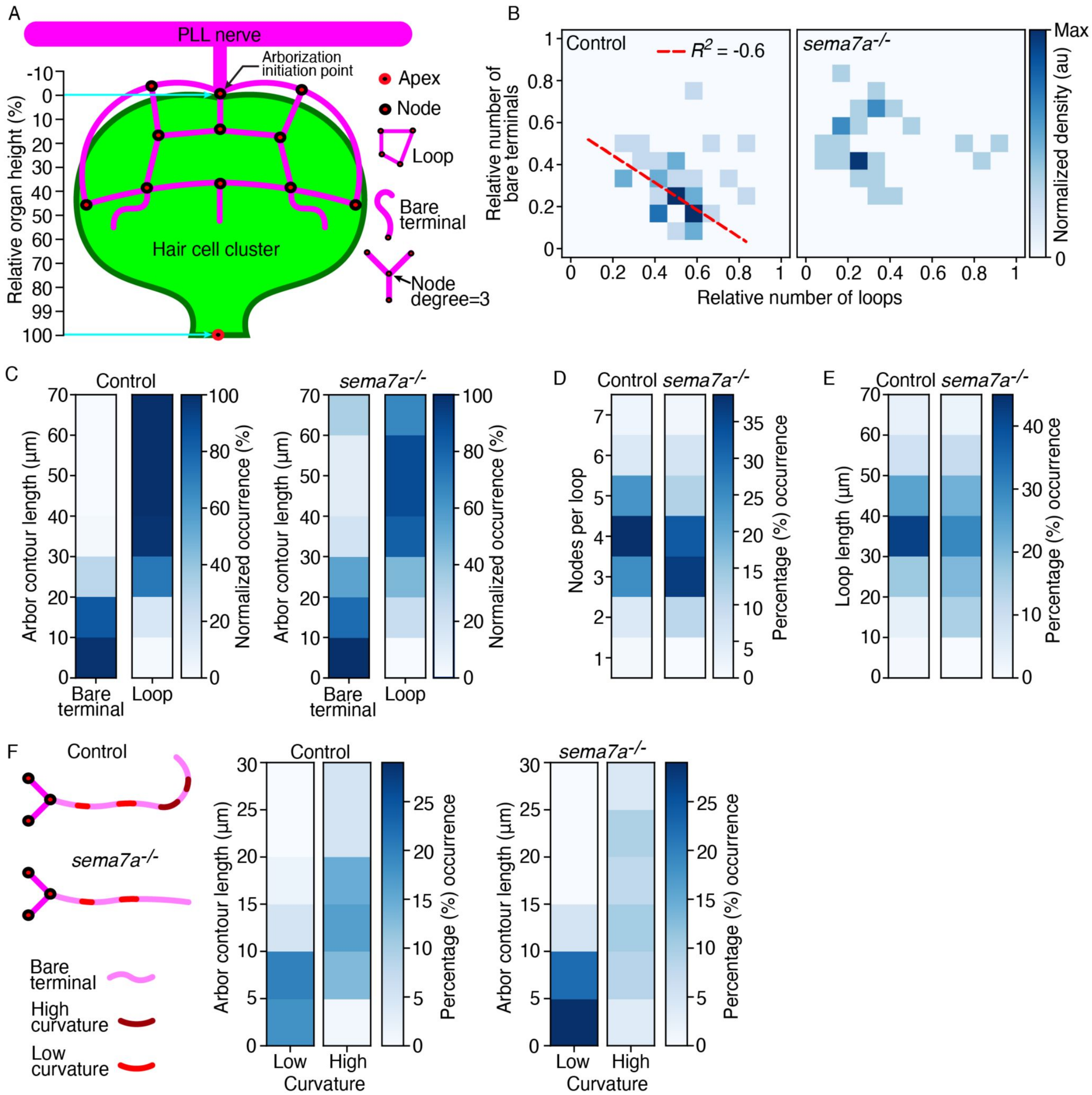

Figure 3.

Sema7A patterns the topology of the sensory arbor network.

(A) A schematic drawing of the combined hair cell cluster (green) and associated sensory axons (magenta) shows diverse topological features of the arborization network, including nodes, loops, bare terminals, node degree, and bare-terminal curvature. The apex is defined as the center of the hair bundle cluster and the base is denoted by the node from which the arborizations extends. The relative organ height describes the position along the axis formed between the arborization initiation point and the apex of the hair cell cluster. (B) The correlation between the relative number of bare terminals and loops from 35 control neuromasts and 27 *sema7a*$^{-/-}$ neuromasts at 4 dpf shows a negative correlation in the control but a scattered distribution in the mutants. The same neuromasts were analyzed in the subsequent panels. R^2, correlation coefficient. Dashed red line depicts the regression slope. (C) The relationship of contour lengths to bare terminals and loops reveals reduced looping in the mutants. (D) Quantification of the number of nodes per loop demonstrates a decrease in mutants. (E) Mutants in the same larvae manifest a broader distribution of loop lengths. (F) A schematic diagram of the extending bare terminals depicts regions of high and low curvature along the length of the arbor. The distributions of bare terminals between low-and high-curvature groups show an enrichment of long, highly curved arbors in the control and short, linear arbors in the mutants. Statistical analyses are displayed in **Table 1**.

Height of node degrees	p-values (control vs *sema7a*$^{-/-}$)		
	2 dpf	3 dpf	4 dpf
Degree = 3	3.50E-01	1.05E-08***	1.68E-03**
Degree = 4	3.85E-01	7.11E-02	1.71E-03**
Degree = 5	4.60E-01	5.23E-01	4.40E-01
Bare terminals vs Loop	2 dpf	3 dpf	4 dpf
0 to 10 μm	3.84E-01	7.89E-01	2.11E-01
10 to 20 μm	3.50E-02*	3.92E-01	2.94E-02*
20 to 30 μm	2.28E-04***	2.63E-02*	3.82E-09***
30 to 40 μm	5.04E-59***	3.63E-25***	1.28E-33***
40 to 70 μm	—	—	—
Nodes per loop	5.06E-01	1.96E-01	3.50E-02*
Loop length	5.81E-01	2.71E-01	2.21E-02*
Distribution of bare terminal curvature	2 dpf	3 dpf	4 dpf
	4.23E-01	8.91E-01	1.99E-02*

Table 1.

Statistical comparison between control and *sema7a*$^{-/-}$ larvae for topological features.

One-sided Kolmogorov-Smirnov tests were used to assess the difference in node degree heights, nodes per loop, and loop lengths for control *versus sema7a*$^{-/-}$ neuromasts at 2 dpf, 3 dpf, and 4 dpf. In each of these cases, the difference in the distributions were not significantly different (p-values in red) between the control and the mutants at 2 dpf and 3 dpf but became significantly different (p-values in green) at 4 dpf. Chi-square tests were used to assess the difference in occurrences of bare terminals versus loops, as well as the occurrences of bare terminals with low versus high curvature. The distribution at 4 dpf shows significant difference between the control and the mutant larvae. The occurrence of significant differences at 4 dpf for diverse topological features suggest that Sema7A function is required to achieve an ordered network topology at a mature stage. Thirty-three, 29, and 35 neuromasts were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively, and 17, 53, and 27 neuromasts were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$ mutant larvae, respectively.
* Signifies $p < 0.05$, ** signifies $p < 0.01$, and *** signifies $p < 0.001$.

perturbed in the *sema7a*$^{-/-}$ larvae across developmental stages, as many of the sensory arbors longer than 30 µm remained as bare terminals (**Figure 3C**; **Figure 3—figure supplement 1F**,**G**; **Table 1**). Furthermore, in 4 dpf control neuromasts the majority of the loops contained four nodes and utilized 30-40 µm of arbors to form closed cycles. These conserved topological attributes were significantly perturbed in the *sema7a*$^{-/-}$ larvae of same age (**Figure 3D**, **E**; **Figure 3—figure supplement 2A-D**; **Table 1**). These findings indicate that Sema7A tightly monitors the sensory arbor loop formation, including the preferred loop length and number of nodes per loop, to properly surround and contact the hair cell cluster.

As observed previously, bare terminals in control neuromasts project transverse to the organ and linger nearby, often curving back to form new loops and contact the hair cell cluster (**Figure 2D**,**F**). To analyze the propensity of the bare terminals to curl backward, we measured their total curvatures, defined as the sum of the curvature at each point along the length of an arbor. Henceforth, we address the total curvature as the curvature of a bare terminal. We classified the bare terminals in two categories: high-curvature arbors with curled morphology and curvature greater than $\pi/6$ (30 degrees); and low-curvature arbors with linear morphology and curvature less than π /6. In control neuromasts 44.8 % of arbors had low curvature compared to 56.4 % in *sema7a*$^{-/-}$ neuromasts at 4 dpf. Concomitantly, 55.2 % of arbors had high curvature in control neuromasts compared to 43.6 % in *sema7a*$^{-/-}$neuromasts at 4 dpf (**Figure 3F**; **Figure 3—figure supplement 2E**, **F**; **Table 1**). These observations further confirm our hypothesis that a Sema7A-mediated attractive cue is involved in orienting bare neural arbors toward the hair cell cluster to potentially form loops.

In summary, we propose that Sema7A mediates four key aspects of the topology of the arborization network: (1) the node degree complexity along the organ's axis; (2) the transition of arbors from bare terminals to loops; (3) the preference for forming loops containing four nodes with 30-40 µm of arbors; and (4) the preference of bare terminals to adopt higher curvature.

Sema7A^{sec} is a sufficient chemoattractive cue for sensory axon guidance

Because genetic inactivation of the *sema7a* gene disrupts both signaling modalities of the cognate protein, we could not assess the specific activity of the Sema7A^{sec} diffusive cue in guiding lateral-line sensory axons. To independently verify the potential role of that isoform, we therefore expressed the secreted variant ectopically and investigated the resultant arborization patterns.

Analysis of the microcircuit connectivity of the neuromasts has demonstrated bare sensory-axonal terminals in the perisynaptic compartments, where they do not contact the hair cell membrane (8). We speculate that these bare terminals are attracted by a Sema7A^{sec} diffusive cue. To test this hypothesis, we ectopically expressed Sema7A^{sec} protein tagged with the fluorophore mKate2, then analyzed its effect on the morphology of sensory arbors. Fertilized one-cell embryos were injected with the *hsp70:sema7a*^{sec}*-mKate2* plasmid and raised to 3 dpf. Larvae expressing the transgenesis marker—the lens-specific red fluorescent protein mCherry—were heat-shocked, incubated to allow expression, and subsequently mounted for live imaging. In these larvae, a random mosaic of cells—often embryonic muscle progenitors in the dermomyotome or mature myofibers—expressed the exogenous Sema7A^{sec}-mKate2 protein. We selected only those neuromasts in which a single mosaic ectopic integration had occurred near the network of sensory arbors (**Figure 4A**,**B**). In all 22 such cases, we observed robust axonal projections from the sensory arbor network toward the ectopically expressing Sema7A^{sec} cells. When the exogenous *sema7a*^{sec} integration occurred in embryonic muscle progenitor cells, which reside in a superficial layer external to the muscle fibers and adjacent to the larval skin (30), the projections were able to form direct contacts with them (**Figure 4C**; Videos 5,6). But more often, when mature myofibers expressed the exogenous Sema7A^{sec} deep in the myotome, the axonal projections approached those targets but failed to contact them (**Figure 4D**, Videos 7,8). This behavior likely arose because other components in the myotomal niche—including the fibrous epimysium, other myofibers, myoblasts, and immature myotubes whose surroundings are permeable to diverse diffusive cues (31)—physically obstructed direct interaction of the extended neurites with the ectopically expressing myofibers. All 18 of the injected but not heat-shocked control larvae did not express ectopic Sema7A^{sec}, and we did not observe aberrant projections from the sensory arbor network (**Figure 4E**).

To measure the accuracy of the extended axonal projections in finding the ectopic Sema7A^{sec} source, we defined three parameters: (1) the source path, denoted as the linear distance from the point of arborization to the boundary of an ectopically expressing Sema7A^{sec} cell; (2) the projection path, representing the linear distance from the point of arborization to the terminus of the extended projection; and (3) the projection proximity, calculated as the linear distance from the terminus of the extended axonal projection to the nearest boundary of the ectopically expressing Sema7A^{sec} cell (**Figure 4B**). We quantified these parameters from 18 of the 22 mosaic integration events; in the remainder the extended axonal projections– although following the ectopic source–reentered the posterior lateral-line nerve so that identification of the axon terminals was not possible (**Figure 4—figure supplement 1A-A**”; Video 9). When we plotted the projection path length against the source-path length, we observed a nearly perfect correlation (**Figure 4F**). This result signifies that irrespective of the location of the ectopic source–whether proximal or distal to the sensory arbor network–the Sema7A^{sec} cue sufficed to attract axonal terminals from the sensory arbor. Furthermore, the low average projection-proximity length of 3.5 ± 0.7 µm (mean ± SEM) indicated that the extended projection terminals remained either in contact (8 of 18) or in the vicinity (10 of 18) of the ectopic Sema7A^{sec} sources (**Figure 4G**). These observations together demonstrate that the Sema7A^{sec} diffusive cue is sufficient to provide neural guidance *in vivo*.

As the posterior lateral-line ganglion matures, newly formed neurons extend their axonal growth cones along the posterior lateral-line nerve to reach the neuromasts (32). We wondered whether the axonal projections that had yet to be associated with a neuromast could also respond to an ectopic source of Sema7A^{sec}. Indeed, on seven occasions in which *sema7a^{sec}* integration had occurred between two neuromasts, we detected neurite extensions from the posterior lateral-line nerve to the ectopic Sema7A^{sec} sources (**Figure 4H**; **Figure 4—figure supplement 1B-B**”, Videos 10-12). Live time-lapse video microscopy further revealed dynamic association between the extending neurites and the ectopic Sema7A^{sec} sources. Long and thin along their entire extents, the neurites were highly motile, extending and retracting rapidly near the Sema7A^{sec} expressing cells. Many neurites maintained their contact with the ectopic sources of Sema7A^{sec} from one hour to almost eight hours (**Figure 4I**,**J**; **Figure 4—figure supplement 2A**, Videos 13-16). The responsiveness of the posterior lateral-line axons to Sema7A^{sec} might therefore be an intrinsic property of the neurons that does not require the neuromast microenvironment.

Sema7A-GPI fails to impart sensory axon guidance from a distance

We speculate that the GPI-anchored transcript variant of the *sema7a* gene can function only through contact-dependent pathways. To test this hypothesis, we expressed the Sema7A-GPI ectopically and investigated the resultant arborization patterns of the sensory axons. Fertilized one-cell embryos were injected with the *hsp70:sema7a-GPI-2A-mCherry* plasmid and raised to 3 dpf. Larvae expressing the transgenesis marker—the lens-specific green fluorescent protein (GFP)—were heat-shocked, incubated to allow expression, and subsequently mounted for live imaging (**Figure 5A**). In these larvae, a few mature myofibers expressed the exogenous Sema7A-GPI protein. We selected only those neuromasts in which an ectopic integration had occurred near the network of sensory arbors. In all 15 such cases, no sensory axon left the arbor network to approach the ectopic Sema7A-GPI sources. In all 17 of the uninjected and heat-shocked control

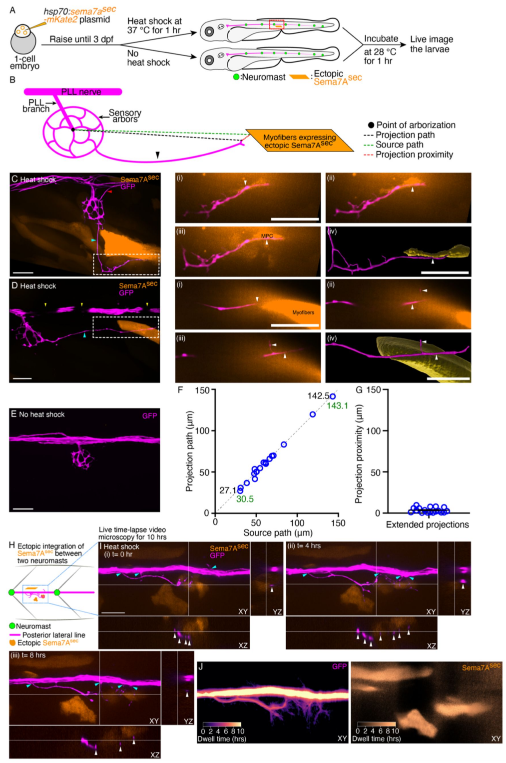

Figure 4.

Ectopic Sema7A^{sec} diffusive cue provides neural guidance *in vivo*.

(A) A diagrammatic overview depicts the generation of a transgenic animal that expresses the Sema7A^{sec} protein ectopically under the control of a thermally inducible promoter. Larvae with ectopic myotomal-integration near the network of sensory arbors (red rectangle) were imaged to analyze arbor morphology. (B) A schematic drawing of a sensory arbor from a heat-shocked larva depicts an extended axonal projection (arrowhead) that reaches toward the myofibers expressing the ectopic Sema7A^{sec} protein. Parameters that quantitate the accuracy of the extended axonal projections toward the ectopic Sema7A^{sec} sources are denoted. (C) In a micrograph of an ectopically expressing Sema7A^{sec} (orange) larva, the sensory arbor (magenta) extends two aberrant axonal projections. One elongates (cyan arrowhead) along the somite boundary to reach and contact an ectopically integrated muscle progenitor cell (white dashed line) and the other (red arrowhead) reenters the posterior lateral-line nerve while following a second ectopic source. The through-focus scans (i–iii) from the epidermis to the dermomyotome and the three-dimensional (3D) surface reconstruction (iv) reveal the intimate contact between the aberrant sensory arbor (arrowheads) and the muscle progenitor cell. (D) In a micrograph of an ectopically expressing Sema7A^{sec} (orange) larva, the sensory arbor (magenta) extends a single aberrant axonal process (cyan arrowhead) to reach ectopically integrated myofibers (white dashed line). The through-focus scans (i–iii) from the epidermis to the dermomyotome and the 3D surface reconstruction (iv) reveal the proximal association of the aberrant sensory arbor (arrowheads) to the myofibers. Melanocytes (yellow arrowhead) along the horizontal myoseptum intermittently block the visibility of the lateral-line nerve. (E) An injected, but not heat-shocked, control larva does not express ectopic Sema7A^{sec} and does not show aberrant projection from the sensory arbor. (F) A plot demonstrates the accuracy of 18 extended axonal arbors in finding ectopic Sema7A^{sec} sources. Each circle represents a single ectopic integration event. The two pairs of numbers represent the minimal and maximal lengths of the projection path (black) and its corresponding source path (green). (G) A plot quantitates the distribution of projection-proximity length from 18 ectopic integration events. (H) A schematic drawing of a section of the lateral-line nerve between two neuromasts from a heat-shocked larva depicts a few extended neurites (magenta) reaching toward the cells expressing the ectopic Sema7A^{sec} protein (orange). The Sema7A^{sec} integration site (blue rectangle) with the sensory axonal projections was imaged for ten hours to visualize axonal dynamics. (I) Micrographs (i-iii) at three distinct times in the time-lapse video microscopy show directed branching (cyan arrowheads) of the sensory neurites from the lateral-line nerve toward the ectopic Sema7A^{sec} sources. Each panel depicts the ectopic integration site and the associated sensory neurites from three different planes, XY, XZ, and YZ. White arrowheads in XZ and YZ planes highlight the close association of the sensory axons with the ectopic Sema7A^{sec} sources. (J) The micrographs depict in pseudocolored trajectories the dwell times of the sensory neurites (left) at the Sema7A^{sec} sources (right) for ten hours. MPC, muscle progenitor cell; scale bars, 20 μm; means ± SEMs.

larvae, the sensory arbor network did not show any perturbation (**Figure 5B-C**, Videos 17-19). These data suggest that the Sema7A-GPI, unlike its secreted counterpart, functions at the surfaces of the expressing cells.

Sema7A deficiency impacts synaptic assembly

Contact stabilization between an axon and its target cell is essential for synapse formation (33). Neural GPI-anchored proteins can act as regulators of various synaptic-adhesion pathways (34). In the mouse olfactory system, GPI-anchored Sema7A is enriched in olfactory sensory axons and mediates sustained interaction with the dendrites of mitral and tufted cells in the olfactory bulb to establish synapses (15). In the developing mouse brain–particularly in Purkinje cells–GPI-anchored Sema7A instead regulates the elimination of synapses onto climbing fibers (35). The role of anchored Sema7A in regulating synaptic architecture can thus vary according to the cell type and developmental stage. Our demonstration that Sema7A is necessary to consolidate the contact between sensory axons and hair cells during neuromast development suggests a possible role in regulating synaptic structure.

Proper synaptogenesis requires the correct spatial organization of the presynaptic and postsynaptic apparatus at the apposition of two cells. At the interface between hair cells and sensory axons, the synaptic network involves two scaffolding proteins in particular: ribeye, a major constituent of the presynaptic ribbons (36), and membrane-associated guanylate kinase (MAGUK), a conserved group of proteins that organize postsynaptic densities (37). Utilizing these scaffolding proteins as markers, we investigated the impact of the *sema7a*$^{-/-}$ mutation on synapse formation. To analyze the distribution of presynaptic densities–immunolabeled with an antibody against the ribeye-associated presynaptic constituent C-terminal binding protein (CTBP) (36)–we counted the number and measured the area of CTBP punctae from multiple neuromasts at several stages of neuromast development (**Figure 6A**,**B**). The CTBP punctae in each hair cell ranged from zero to seven across developmental stages. In the zebrafish's posterior lateral line, the formation of presynaptic ribbons is an intrinsic property of the hair cells that does not require contact with lateral-line sensory axons. However, sustained contact with the sensory axon influences the maintenance and stability of ribbons (38). As neuromasts matured from 2 dpf to 4 dpf, we identified similar distribution patterns with a characteristic increase in the number of CTBP punctae in both control and *sema7a*$^{-/-}$larvae (**Figure 6C**,**D**). This result implies that the formation of new presynaptic ribbons is not perturbed in the mutants, even though there are on average fewer CTBP punctae (**Figure 6—figure supplement 1A-C**). Measurement of the areas of CTBP punctae also showed similar distributions in both genotypes (**Figure 6E**,**F**) but the average area of presynaptic densities was reduced in the mutants (**Figure 6—figure supplement 1D-F**, **Figure 6—figure supplement 2**). Because GPI-anchored Sema7A lacks a cytosolic domain, it is unlikely that Sema7A signaling directly induces the formation of presynaptic ribbons. We propose that the decrease in average number and area of CTBP punctae likely reflects decreased stability of the presynaptic ribbons owing to lack of contact between the sensory axons and the hair cells.

Our analysis of the skeletonized network traces of the sensory arbor network identified distinct topological features that are resultant of collective assemblies of each sensory axon. Because postsynaptic aggregates are organized at the sensory axons, we wondered whether they are associated with a specific topological feature of the network. To identify the distribution of the postsynaptic aggregates along the arbor network, we utilized the transgenic animal *TgBAC(neurod1:EGFP)* that labeled the sensory axons of the posterior lateral line in green fluorescence protein (GFP), and marked the postsynaptic aggregates through immunostaining with a pan-MAGUK antibody (**Figure 6G**). We generated skeletonized network traces of the arbor network and determined the three-dimensional spatial locations of the MAGUK punctae (**Figure 6—figure supplement 2**). Comparison of the topological features of the network and the spatial locations of the MAGUK punctae revealed that the postsynaptic densities occur more extensively

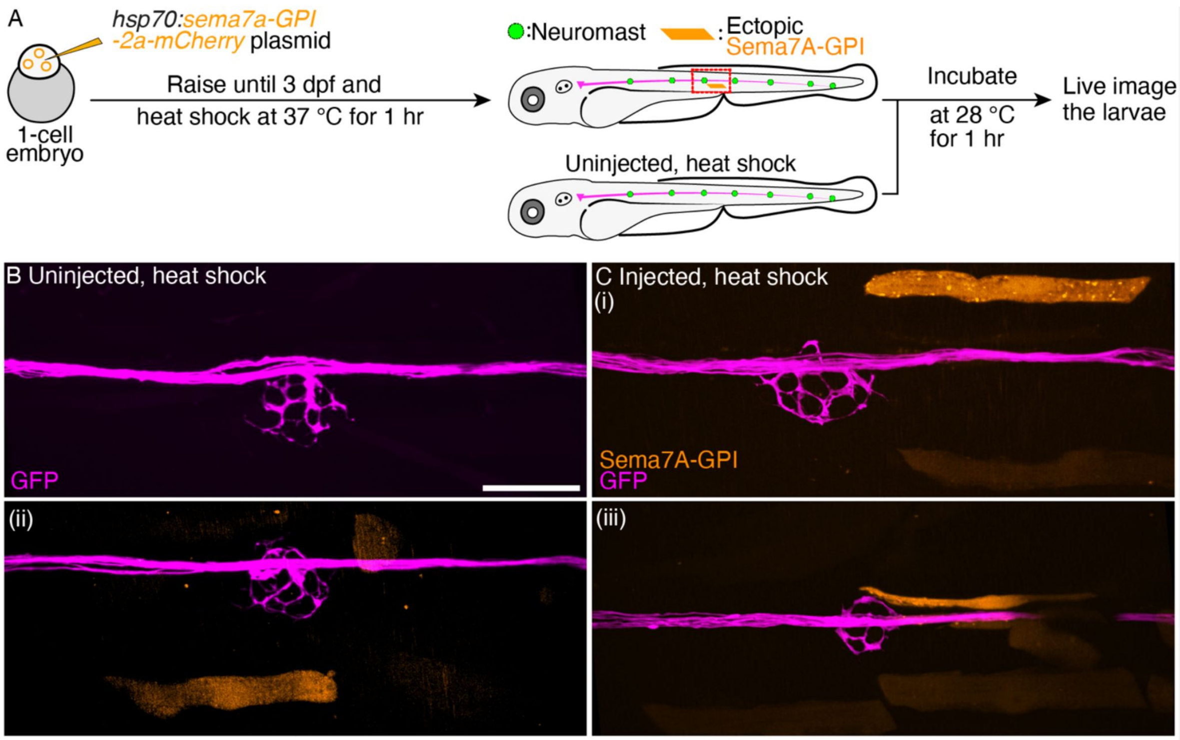

Figure 5.

Ectopic Sema7A-GPI fails to guide sensory arbors from a distance.

(A) A diagrammatic overview depicts the generation of a transgenic animal that expresses the Sema7A-GPI protein ectopically under the control of a thermally inducible promoter. Larvae with ectopic myotomal-integration near the network of sensory arbors (red rectangle) were imaged to analyze arbor morphology. Uninjected, heat-shocked larvae were used as controls. (B) Uninjected and heat-shocked larva depicts normal arborization pattern of the sensory axons (magenta). (C) Injected and heat-shocked larvae robustly express the Sema7A-GPI in the myofibers (orange). Panels (i-iii) depict the ectopically expressing Sema7A-GPI myofibers that fail to attract sensory arbors toward itself. Scale bar, 20 μm.

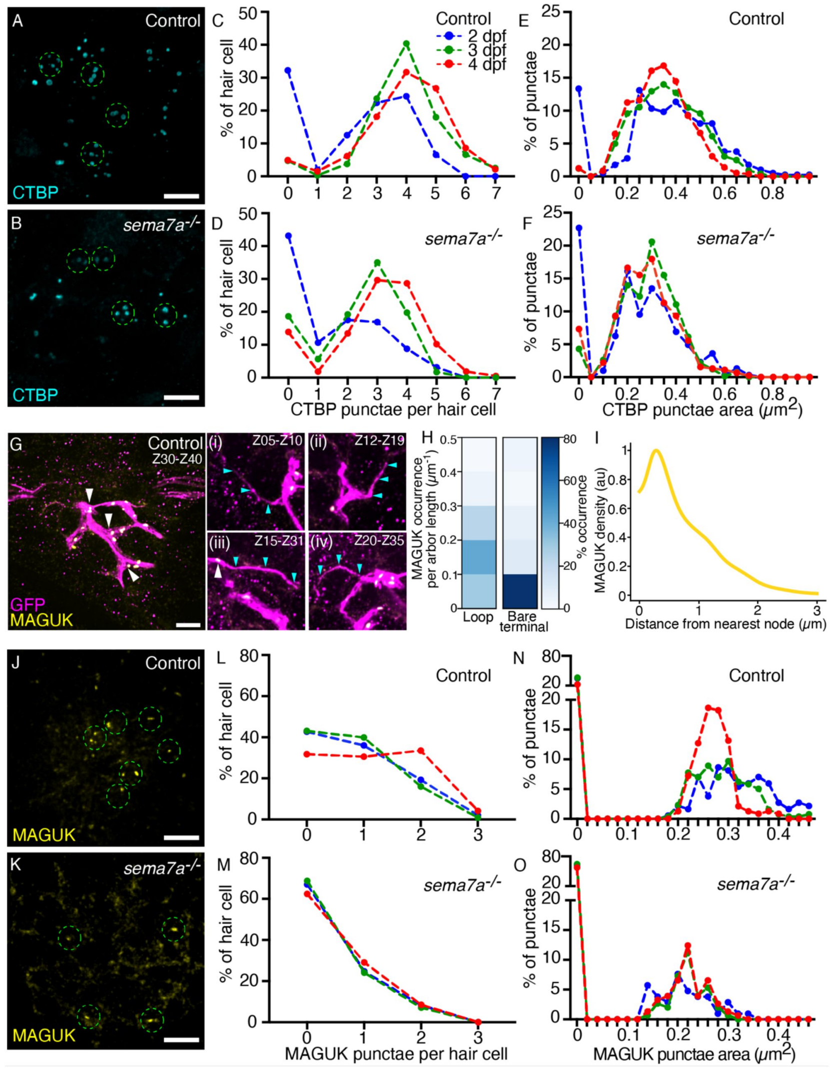

Figure 6.

Sema7A deficiency impairs synaptic assembly.

(A,B) Maximal-intensity projections of micrographs from a control and a *sema7a*$^{-/-}$ neuromast depict the presynaptic aggregates marked by CTBP (cyan). The approximate hair cell basal region is outlined by dashed green circles. (C-F) Plots quantitate the numbers of presynaptic aggregates (C,D) and their areas (E,F), in each hair cell across developmental stages. One hundred and fifty-two, 317, and 325 hair cells were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively. One hundred and sixty, 216, and 177 hair cells were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$mutant larvae, respectively. (G) A micrograph generated from maximal-intensity projection of ten consecutive through-focus scans across Z-planes of a control 4 dpf neuromast depict the sensory arbors (magenta) and the postsynaptic aggregates marked by MAGUK (yellow, white arrowheads). The MAGUK punctae primarily occur on the loop and aggregate near the nodes. Panels (i-iv) depict maximal-intensity projection of consecutive through-focus scans of distinct Z-planes to show bare terminals. In each panel, the bare terminal (magenta, cyan arrowheads) rarely harbors postsynaptic aggregates (yellow, white arrowhead). (H) Quantification of the distribution of the MAGUK-punctae between the two topological features of the network depicts enrichment of postsynaptic densities on the loops over the bare terminals. (I) A histogram shows the distribution of MAGUK-punctae density against the distance from the node. The histogram is represented as a density trace graph (yellow). For both H and I, five hundred and sixty-seven MAGUK punctae were analyzed from 16 neuromasts of 4 dpf larvae. (J,K) Maximal-intensity projections from micrographs of a control and a *sema7a*$^{-/-}$ neuromast depict the postsynaptic aggregates marked by MAGUK (yellow). The approximate hair cell basal region is outlined by dashed green circles. (L-O) Plots quantitate the numbers of postsynaptic aggregates (L,M) and their areas (N,O), in each hair cell across developmental stages. One hundred and fifty, 218, and 167 hair cells were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively. Ninety-seven, 141, and 141 hair cells were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$ mutant larvae, respectively. Scale bars, 5 µm.

on the loops than on the bare terminals (**Figure 6G**,**H**). We measured 567 MAGUK punctae from 16 sensory arbor networks, where 531 punctae were located on the loops and only 36 punctae on the bare terminals. Furthermore, the MAGUK punctae predominantly aggregate near the node and their density sharply diminishes moving farther from the node (**Figure 6I**, **Figure 6—figure supplement 2**). These observations suggest that the distribution of the postsynaptic aggregates is closely associated with the topological attributes of the sensory circuit. Because in the *sema7a*$^{-/-}$ mutants the arbor networks have significantly lower number of loops and nodes, and more bare terminals, we hypothesize that in the mutants the occurrence of postsynaptic aggregates is perturbed. Moreover, Sema7A-GPI-mediated juxtracrine signaling onto the apposed axonal membrane is essential to induce postsynaptic assembly (15).

To characterize the impact of *sema7a* mutation on the formation of postsynaptic aggregates, we counted the number and measured the area of MAGUK punctae from multiple neuromasts at several stages of neuromast development (**Figure 6J**,**K**). At 4 dpf, 68.3 % of the hair cells in the control neuromasts were associated with at least one MAGUK punctum. In the *sema7a*$^{-/-}$ mutant this value fell to 37.5 % (**Figure 6L**,**M**). The average number of MAGUK punctae also declined in the mutants (**Figure 6—figure supplement 1G-I**). As control neuromasts matured from 2 dpf to 4 dpf, the MAGUK punctae showed a shift toward smaller sizes, suggesting that the postsynaptic structure fragmented into smaller entities as observed in other excitatory synapses (40). At 4 dpf, 77.5 % of the hair cells in the control neuromasts had postsynaptic densities with areas between 0.10 μm^2 and 0.40 μm^2. In the *sema7a*$^{-/-}$ mutant the corresponding value was only 42.5 % (**Figure 6N**,**O**). The average area of MAGUK punctae was also reduced in the mutants compared to the controls (**Figure 6—figure supplement 1J-L**, **Figure 6—figure supplement 2**). We speculate that the abnormalities in the postsynaptic structure of the *sema7a*$^{-/-}$mutants arose from reduced contact between the hair cells and the sensory axons and perturbed topological attributes of the sensory circuit or from the lack of Sema7A-GPI-mediated juxtracrine signaling onto the apposed neurite terminals (15).

Discussion

The specification of neural circuitry requires precise control of the guidance cues that direct and restrict neural arbors at their target fields and facilitate synapse formation. In the lateral-line neuromast of the zebrafish, we have identified Sema7A as an essential cue that regulates the interaction between hair cells and sensory axons. We have discovered dual modes of Sema7A function *in vivo*: the chemoattractive diffusible form is sufficient to guide the sensory arbors toward their target, whereas the membrane-attached form likely participates in sculpting accurate neural circuitry to facilitate contact-mediated formation and maintenance of synapses. Our results suggest a potential mechanism for hair cell innervation in which a local Sema7A^{sec} diffusive cue likely consolidates the sensory arbors at the hair cell cluster and the membrane-anchored Sema7A-GPI molecule guides microcircuit topology and synapse assembly.

The role of Sema7A in regulating diverse topological features of the arborization network suggests that microcircuits of control and *sema7a*$^{-/-}$neuromasts have important functional differences. Loss of Sema7A disrupts the hierarchical organization of the node-degree distribution along the organ's axis, as well as the preference towards forming loops containing four nodes with 30-40 μm of arbors. What benefit does the network gain from surrounding the hair cell cluster with closed loops instead of bare terminals? The network may actively form loops to contribute to the neuromast's overall structure and function, or else emerge as a consequence of the arbors navigating the intervening spaces between the hair cells and support cells. The preferential localization of the postsynaptic aggregates in the loops and an enrichment near the nodes of the arbor network suggest that the precise patterning of the network—with characteristic looping behaviors—is not merely a byproduct of neuromast architecture, but may contribute to the network's ability to gather sensory inputs through properly placed postsynaptic aggregates and to faithfully transmit sensory information to higher processing centers in the hindbrain (7). Further studies are necessary to elucidate the potential role of these topological features in regulating the distribution of synaptic bodies and optimizing robust neural transmission in the lateral-line microcircuit.

Recent studies have begun to dissect the molecular mechanisms of semaphorin signaling in axonal navigation. In explants derived from the murine olfactory bulb, Sema7A interacts directly with integrinβ1 to trigger downstream signaling effectors involving focal adhesion kinase and mitogen-activated protein kinases, which are critical regulators of the cytoskeletal network during axonal pathfinding (4,41). Moreover, the interaction between Sema7A and its receptor plexinC1 activates the rac1-cdc42-PAK pathway, which alters the actin network to promote axonal guidance and synapse formation (15,42). We speculate that lateral-line sensory axons utilize similar mechanisms to sense and respond to Sema7A in the establishment of microcircuits with hair cells.

During early development of the posterior lateral line, the growth cones of sensory axons intimately associate with the prosensory primordium that deposits neuromasts along the lateral surface of the larva (43,44). The growth of the sensory fibers with the primordium is regulated by diverse neurotrophic factors expressed in the primordium (45,46). Impairing brain-derived or glial cell line-derived neurotrophic factors and their receptors perturbs directed axonal motility and innervation of hair cells (45,46). Lateral-line hair cells are particularly enriched in brain-derived neurotrophic factor (47) and other chemotropic cues (**Figure 7**). Because the sensory axons of *sema7a*$^{-/-}$ mutants reliably branch from the lateral-line nerve to reach hair cell clusters, these chemoattractive factors might compensate in part for the absence of Sema7A function. Although sensory axons are strict selectors of hair cell polarity and form synapses accordingly (5), diverse chemoattractant cues including Sema7A occur uniformly between hair cell polarities (47). It seems likely that unbiased attractive signals bring sensory arbors toward hair cells, after which polarity-specific molecules on hair cell surfaces dictate the final association between axonal terminal and hair cells. Notch-delta signaling and its downstream effectors regulate hair cell polarity and the innervation pattern (6,8,27). It would be interesting to determine whether Notch-mediated polarity signatures on the hair cell's surface can refine the neuromast neural circuitry.

Materials and Methods

Zebrafish strains and husbandry

Experiments were performed on 1.5-4 dpf zebrafish larvae in accordance with the standards of Rockefeller University's Institutional Animal Care and Use Committee. Naturally spawned and fertilized eggs were collected, cleaned, staged, and maintained at 28.5 °C in system-water with 1 μg/mL methylene blue. Wild-type AB and heterozygous mutant *semaphorin7a*^{sa24691} zebrafish were obtained from the Zebrafish International Resource Center. In addition, the following transgenic zebrafish lines were used: *Tg(myo6b:actb1-EGFP)* (48), *Tg(neurod1:tdTomato)* (27), *Tg(cldnb:LY-mScarlet)* (26), and *TgBAC(neurod1:EGFP)* (39).

Embryo dissociation and flow cytometry

Immediately before dissection, 4-day-old *Tg(myo6b:actb1-EGFP)* and wild-type larvae were anesthetized in 600 μM 3-aminobenzoic acid ethyl ester methanesulfonate in system water with 1 μg/mL methylene blue. The anterior half of each larva was removed with a fine surgical blade. We used approximately 400 GFP-positive and 50 GFP-negative larvae, the latter for controls of the gating procedure. Dissociation of the tails were performed using a protocol (22) with minor modification; detailed protocol is available upon request. The resuspended cells were incubated for 5 min on ice and sorted with an influx cell sorter (BD Biosciences, San Jose, CA, USA) with an 85

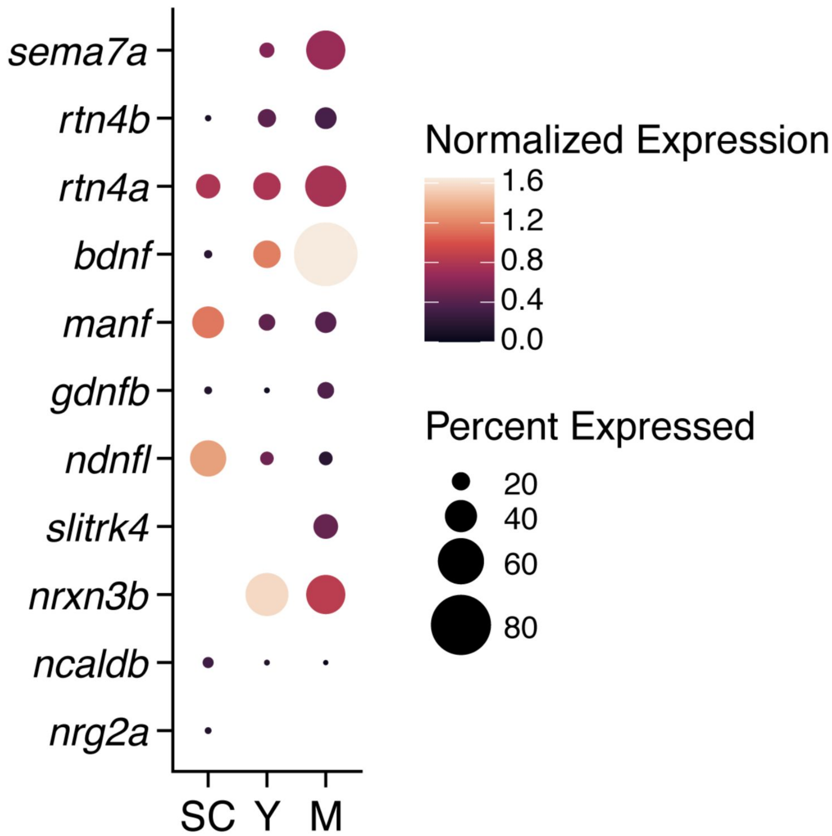

Figure 7.

Expression of diverse neural guidance cues in developing neuromasts.

Analysis of single cell RNA-sequencing data (9) shows the expression of multiple neural guidance genes that are expressed across young hair cells (Y), mature hair cells (M), and support cells (SC). Dot sizes and colors signify the proportion of cells in each cluster that express a gene and the average strength of expression (log[(counts/10,000) +1]), respectively.

μm nozzle at 20 psi and 1X DPBS (Sigma-Aldrich, St Louis, USA). Laser excitation was conducted at 561 nm and 488 nm. Distinct populations of GFP-positive hair cells were isolated based on forward scattering, lateral scattering, and the intensity of GFP fluorescence.

RNA extraction and cDNA library preparation

Sorted cells were collected in a lysis-buffer solution supplemented with β-mercaptoethanol and RNA extraction was performed immediately using a standard protocol (RNeasy Plus Micro Kit; Qiagen). The yield and quality of the product were measured with a bioanalyzer (Agilent 2100). Only samples with an RNA integrity score (RIN) greater than 8.0 were selected for the preparation of cDNA libraries using iScript cDNA Synthesis Kit (BioRad). One nanogram of total RNA was amplified to several micrograms of cDNA.

Reverse transcription-polymerase chain reactions (RT-PCR)

The *sema7a* gene expresses two transcript variants: variant 1, *sema7a-GPI* anchored, NM_001328508.1; and variant 2, *sema7a*^{sec}secreted, NM_001114885.2. The variant 2 lacks several C-terminal exons, and its 3’-terminal exon extends past a splice site that is used in variant 1. The encoded variant 2 has a shorter and distinct C-terminus than variant 1. To individually identity these transcripts, each of them was PCR amplified from the cDNA library using the following primers: For variant 1, sema7aF= 5'-GGTTTTTCTGAGGCCATTCC-3', sema7aR1= 5'-GGCACTCGTGACAAATGCTA-3'; and for variant 2, sema7aF= 5'-GGTTTTTCTGAGGCCATTCC-3', sema7aR2= 5'-TGTGGAGAAAGTCACAAAGCA-3'. To detect *pvalb8* transcript we used the following primers: pvalb8F= 5'-ATGTCTCTCACATCTATCCT-3' and pvalb8R= 5'-TCAGGACAGCACCATCGCCT-3'. To detect *s100t* transcript we used the following primers: s100tF= 5'-ATGCAGCTGATGATCCAGAC-3' and s100tR= 5'-CTATTTCTTGTCATCCTTCT-3'.

Immunofluorescence microscopy

For the immunofluorescence labeling of wholemounted wild-type control and *sema7a*$^{-/-}$ larvae, 1.5-4 dpf larvae were fixed overnight at 4 °C in 4 % formaldehyde in phosphate-buffered saline solution (PBS) with 1 % Tween-20 (1 % PBST). Larvae were washed four times with 1 % PBST for 15 min each, then placed for 2 hr in blocking solution containing PBS, 2 % normal donkey serum (NDS), 0.5 % Tween-20, and 1 % BSA. Primary antibodies were diluted in fresh blocking solution and incubated with the specimens overnight at 4 °C. The primary antibodies were goat anti-Sema7A (1:200; AF1835, R&D Systems) (**Figure 1—figure supplement 4**) (14), rabbit anti-myosin VI (1:200; Proteus 25-6791), murine anti-GM130 (1:200; 610822, BD Transduction Laboratories), murine anti-CTBP (1:200; B-3, SC-55502, Santa Cruz Biotech), and rabbit anti-mCherry (1:200; GTX128508, GeneTex). Larvae were washed four times with 0.1% PBST for 15 min each. Alexa Fluor-conjugated secondary antibodies (Invitrogen, Molecular Probes) applied overnight at 1:200 dilutions in 0.2 % PBST included anti-rabbit (488 nm), anti-goat (555 nm), and anti-mouse (647 nm). Larvae were then washed four times in 0.2 % PBST for 15 min each and stored at 4 °C in VectaShield (Vector Laboratories).

For the immunofluorescence labeling of postsynaptic density in wholemounted wild-type control and *sema7a*$^{-/-}$ larvae, 2-4 dpf larvae were fixed with 4 % formaldehyde in PBS for 4.5-6 hr at 4 °C. Larvae were washed five times with 0.01 % PBST for 5 min each and then rinsed in distilled water for 5 min. The larvae were permeabilized with ice-cold acetone at −20 °C for 5 min. They were rinsed in distilled water for 5 min, washed five times with 0.01 % PBST for 5 min each, and blocked overnight with PBS buffer containing 2 % goat serum and 1 % BSA. Murine anti-pan-MAGUK antibody (1:500; IgG1 NeuroMab, K28.86, 75-029 and MABN72, Millipore) was diluted in PBS containing 1 % BSA, added to the larvae, and incubated for 3 hr at room temperature. Larvae were washed five times for 5 min each with 0.01 % PBST. Alexa Fluor 555-conjugated anti-mouse secondary antibody (1:1000) diluted in PBS containing 1 % BSA was added to the larvae and incubated for 2 hr at room temperature. Larvae were washed five times with 0.01 % PBST for 5 min each, rinsed in distilled water for 5 min, and stored at 4 °C in VectaShield (Vector Laboratories).

Posterior segments of fixed larvae were mounted on a glass slide and imaged on an Olympus IX81 microscope with a microlens-based, super-resolution confocal microscope (VT-iSIM, VisiTech international) under 60X and 100X, silicone-oil objective lenses of numerical apertures 1.30 and 1.35, respectively. Static images at successive focal depths were captured at 200 nm intervals and deconvolved with the Microvolution software in ImageJ.

Measurement of Sema7A and td-Tomato^{+} sensory arbor intensities

The average Sema7A fluorescence intensities of the hair cells from 1.5 dpf, 2 dpf, 3 dpf, and 4 dpf neuromasts were measured by determining the mean gray level within each hair cell labeled by actin-GFP expression. The intensity distribution of Sema7A protein across the basal cell membrane was measured using the line profile tool in ImageJ (**Figure 1—figure supplement 2F-H**). The lines were drawn with 1 μm length, starting from the inside of the hair cell to the outside. For each hair cell membrane, the intensity values were scaled from 0 to 1 arbitrary units. The intensity distribution of Sema7A protein along the apicobasal axis of the hair cell was measured using the line profile tool in ImageJ (**Figure 1—figure supplement 2I**). Because hair cell lengths differed among samples, lines were drawn with 2 μm widths and varying lengths of 6-8 μm. For each hair cell, apicobasal length was scaled from 0 to 100 arbitrary units and the intensity values from 0 to 1 arbitrary units. The average Sema7A and td-Tomato^{+} sensory arbor fluorescence intensities at the hair cell bases from 2 dpf, 3 dpf, and 4 dpf neuromasts were measured by determining the mean gray level within each hair cell basal area, denoted by the region underneath the hair cell nucleus (**Figure 1—figure supplement 3A-E**).

Genotyping of mutant fish

The *semaphorin7a^{sa24691}* allele harbors an A-to-T point mutation in the seventh exon of the *semaphorin7a* gene, which creates a premature stop codon (**Figure 2—figure supplement 1A**). Genomic DNA was isolated from the tail fins of adult *semaphorin7a^{sa24691}* (*sema7a^{sa91}*) heterozygous mutant zebrafish. The mutant locus in the genome was PCR amplified using the following primers: sema7a^{sa91}F: 5'-AAAGCTGGAAAGCGAATCAA-3' and sema7a^{sa91}R: 5'-ATATCCAAGGATCCGCCTCT-3'. The 734-base pair amplicons were sequenced, and heterozygous adults were propagated.

Lipophilic styryl fluophore labeling of hair cells

Freely swimming 4 dpf control and *sema7a$^{-/-}$* larvae were exposed to the styryl fluorophore FM4-64 (N-(3-triethylammoniumpropyl)-4-(6-(4-(diethylamino) phenyl) hexatrienyl) pyridinium dibromide, ThermoFischer Scientific) at 5 μM for 30 s, then rinsed three times in fresh embryo water and immediately mounted in low-melting-point agarose for live imaging.

Transient transgenesis with the hsp70l:sema7a^{sec}-mKate2 construct

The pDNR-LIB plasmid containing the *sema7a^{sec}* coding sequence was purchased from horizon (7140389, Perkin-Elmer). The coding sequence was PCR amplified using the following primers: sema7a^{sec}F: 5'-GGGGACAAGTTTGTACAAAAAAGCAGGCTTGATGATTCGACATTATTCT-3' and sema7a^{sec}R: 5'-GGGGACCACTTTGTACAAGAAAGCTGGGTG*CTT*TGTGGAGAAAGTCACAAAGCA-3'. The italicized nucleotides denote the mutated stop codon. The amplicon was then cloned into the pDONR221 vector by BP recombination to generate the middle entry pME-sema7a^{sec} vector. To generate the hsp70:sema7a^{sec}-mKate2 construct, gateway cloning was performed by combining the plasmids p5E-hsp70 (49), pME-sema7a^{sec}, p3E-mKate2-myc no-pA (Addgene, 80812), and pDESTtol2pACrymCherry (Addgene, 64023) with LR Clonase II Plus (Invitrogen). Verified constructs (25 ng/µl plasmid DNA) were injected with Tol2 Transposase mRNA (approximately 25 ng/µl) into one-cell embryos. The transiently transgenic larvae— identified by the expression of red fluorescent protein (mCherry) in their lenses—were raised till 3 dpf, heat-shocked in a water bath at 37 °C for 1 hr, incubated at 28 °C for 1 hr to allow expression, and subsequently mounted for live imaging.

Transient transgenesis with the hsp70l:sema7a-GPI-2A-mCherry construct

The *sema7a-GPI* coding sequence without the stop codon (*sema7a-GPI$^{no\text{-}stop}$*) was synthesized using gBlocks HiFi Gene Fragments (Integrated DNA Technologies). The *sema7a-GPI$^{no\text{-}stop}$* coding sequence was then cloned into the pDONR221 vector by BP recombination to generate the middle entry pME-sema7a-GPI$^{\text{no-stop}}$ vector, henceforth denoted as pME-sema7a-GPI. To generate the hsp70:sema7a-GPI-2A-mCherry construct, gateway cloning was performed by combining the plasmids p5E-hsp70 (49), pME-sema7a-GPI, p3E-2A-mcherrypA (Addgene, 26031), and pDESTtol2pACryGFP (Addgene, 64022) with LR Clonase II Plus (Invitrogen). Verified constructs (25 ng/µl plasmid DNA) were injected with Tol2 Transposase mRNA (approximately 25 ng/µl) into one-cell embryos. The transiently transgenic larvae—identified by the expression of green fluorescent protein (GFP) in their lenses—were raised till 3 dpf, heat-shocked in a water bath at 37 °C for 1 hr, incubated at 28 °C for 1 hr to allow expression, and subsequently mounted for live imaging.

Microscopy and volumetric rendering of living neuronal arbors and cells

Living larvae were dechorionated, anaesthetized in 600 µM 3-aminobenzoic acid ethyl ester methanesulfonate in system-water with 1 µg/mL methylene blue, and mounted in 1 % low-melting-point agarose on a glass-bottom MetTek dish at specific stages. The larvae were maintained at 28 °C during imaging in a temperature-controlled stage top chamber (OKO Lab UNO-T). Static images at successive focal depths were captured at 200 nm intervals and deconvolved with Microvolution software in ImageJ (NIH). For time-lapse video microscopy, the ectopic *sema7a^{sec}* integration sites were imaged at intervals of 5 min as Z-stacks acquired with 1.0 µm steps with laser excitation at 488 nm and 561 nm under a 60X, silicone-oil objective lens of numerical aperture 1.30. The three-dimensional datasets were processed and volume-rendered with the surface evolver and filament-tracer tools in Imaris (Bitplane, Belfast, UK). The Imaris workstation was provided by the Rockefeller University Bio-imaging Resource Center.

Generation of pseudocolored trajectories of the dwell times of sensory neurites at the ectopic Sema7A$^{\text{sec}}$ sources

Fluorescent intensity values of the sensory neurites and the ectopic Sema7A$^{\text{sec}}$ sources were measured from the maximal-intensity projections at each time during the time-lapse video microscopy. The intensity values were normalized and binarized before summing across each timepoint for locations across the XY plane. The resulting pseudocolored trajectories represent the dwell times of the neurites and the Sema7A$^{\text{sec}}$ cells through the 10 hr time-lapse video sequence.

Generation of skeletonized networks

The labeled sensory axons were traced in three dimensions with ImageJ's semi-automated framework, SNT (50). Each trace depicts the skeletonized form of the posterior lateral-line nerve, the posterior lateral-line branch, and the point of arborization from which the sensory arbors radiate to contact the hair cells (**Figure 2D**,**E**; **Figure 2—figure supplement 2B**. The three-dimensional pixels or voxels obtained from the skeletonized sensory arbor traces were processed using custom code written in Python for visual representation and quantitative analysis.

Generation of combined hair cell clusters and the corresponding combined skeletonized networks

The hair cell clusters of neuromasts from each developmental stage were aligned by the centers of their apices, which were identified by the hair bundles, and overlayed to generate combined hair cell clusters (**Figure 2—figure supplement 1E**,**F**) using custom Python code. The associated skeletonized sensory-arbor traces at each developmental stage were aligned similarly using custom Python code. The details of the procedures are available upon request and the corresponding code is available on GitHub.

Quantification of sensory arbor distributions around hair cell clusters

The region from the center of the combined hair cell cluster to 60 µm in the post-cluster region was divided into 3600 concentric sections of equal area. Each section, as depicted with a distinct color shade, had an area of π µm^2. Sensory arbor density was defined as Log_{10}[(Area occupied by the arbors in each section)/ (Area of each section)]. The arbor density was then plotted as a histogram against distance from the center of the combined hair cell cluster. Each bin of the histogram represents an area of π µm^2 (**Figure 2—figure supplement 2A**). The histograms were finally represented as density trace graphs with Kernel Density Estimation (KDE) in Python.

Quantification of contact between sensory arbors and hair cell clusters

For each hair cell cluster, the voxels of the sensory arbor traces that were inside or within 0.5 µm —the average neurite radius—of the cluster boundary were denoted as arbors in contact with their corresponding hair cell cluster (**Figure 2—figure supplement 2B**). The percentage of arbors in contact with each hair cell cluster was defined as: (Number of voxels that remained inside the cluster boundary/Total number of voxels in the arbor) X 100. The data were analyzed through an automated pipeline with custom Python code.

Quantification of the sensory arborization network topology

We developed a custom Python script (snt_topology.py; available on GitHub) to convert the skeletonized sensory arborization networks to network graphs (NetworkX, v3.1). We first used a custom ImageJ macro to convert SNT skeletons to csv files, then generated the network graph from the individual arbor paths. As noted previously, we defined five topological attributes: (1) nodes — junctions where arbors branch, converge, or cross; (2) loops—minimal arbor cycles forming topological holes; (3) bare terminals—arbors with free ends; (4) node degree—the number of first-neighbor nodes incident to a node; and (5) arbor curvature—the total curvature of a bare terminal (**Figure 3A**). To enumerate loops and quantify their lengths and node numbers, we computed the minimum cycle basis for each network graph. The relative organ height of each node in a graph was defined in 3D using **Equation (1)**:

$$(\vec{u} \cdot \vec{v}) \;/\; \|\vec{v}\|^2, \qquad (1)$$

where $\vec{u}$ is the vector from the arborization initiation point to a node and $\vec{v}$ is the vector from the arborization initiation point to the hair cell cluster apex. The relative occurrence of nodes (**Figure 3—figure supplement 1A-C**) was normalized by both the number of hair cells per neuromast and the total number of neuromasts, for comparison across ages and genotypes. The relative number of bare terminals and loops (**Figure 3B**; **Figure 3—figure supplement 1D**, **E**) was normalized by the number of hair cells per neuromast and scaled between 0 and 1. The density was then normalized by the total number of neuromasts imaged for each developmental stage and genotype. For **Figure 3C** and **Figure 3—figure supplement 1F and G**, the normalized occurrence was computed as the number of arbors of a given length that formed either a bare terminal or a loop, divided by the total number of arbors of either feature of that length. Lastly, we defined bare terminals as having either high or low curvature (**Figure 3F**; **Figure 3—figure supplement 2E**,**F**) if their absolute total signed curvature [**Equation (2)**] in three dimensions was either higher or lower than $\pi/6$, respectively:

$$\left| \int \frac{dT}{ds} \cdot N(s)\, ds \right|, \qquad (2)$$

where s is the arc-length parameter of the bare terminal, $T(s)$ is the unit tangent vector at each point along the arbor, $N(s)$ is the unit normal vector at each point, and the integral is evaluated from the terminal's branching node to the arbor's tip. For a closed curve forming a full loop, this value will equal 2π. In practice, we first fit B-splines (scipy.interpolate.splprep; SciPy, v1.10.1) in 3D to the skeletons for each bare terminal before evaluating their derivatives (scipy.interpolate.splev) and integrating (scipy.integrate.simpson) over their lengths.

Quantification of aberrant sensory arborization in ectopic Sema7A^{sec} expression

The sensory arbor networks were traced in three dimensions with SNT. The surface of each of the ectopically expressing Sema7A^{sec}-mKate2 cells was represented by a single point that was closest to the terminal point of the guided neurite. Distances between two points were calculated using standard mathematical operations.

Quantification of presynaptic and postsynaptic assembly number and area

Maximal-intensity projections of consecutive through-focus scans encompassing the entire neuromast of control and *sema7a*$^{-/-}$ mutant larvae at 2 dpf, 3 dfp, and 4 dpf were utilized to capture the presynaptic and postsynaptic assemblies. The number of the synaptic densities across genotypes were measured using ImageJ's cell counter framework. Area of the synaptic densities from the maximal-intensity projections were measured using ImageJ's polygon selection tool (**Figure 6—figure supplement 2A-D**).

Quantification of the distribution of postsynaptic aggregates along the loops and the bare terminals of the sensory arbor networks

Four-day-old *TgBAC(neurod1:EGFP)* larvae were fixed and stained for sensory arbors and postsynaptic aggregates marked by GFP and MAGUK, respectively. The GFP-positive sensory arbors were traced in three dimensions with ImageJ's semi-automated framework, SNT. The topological attributes of the arbor network, such as the loop and the bare terminal, were determined as described previously. The localizations of the MAGUK punctae along the three-dimensional arbor network were measured using ImageJ's cell counter framework (**Figure 6—figure supplement 2E**,**F**). We measured 567 MAGUK punctae from 16 sensory arbor networks. Whether the MAGUK-punctae were associated with a loop or a bare terminal, we categorized them in two groups and quantified their distribution across 16 neuromasts.

Quantification of the distribution of postsynaptic-aggregate density along the distance from the nearest node

The three-dimensional space surrounding each node of the arbor network were divided into concentric shells of equal volume (**Figure 6—figure supplement 2E**, **F**). For each node, we computationally generated 100,000 concentric shells that reached a 10 μm radius. We measured the number of MAGUK punctae located within each of those concentric shells from their nearest nodes.

Statistical analysis

Data visualization and statistical analysis were conducted with GraphPad Prism (Version 9) and Python (v3.10). The Mann-Whitney test was used for hypothesis testing unless otherwise noted, and the statistical details are given in the respective figure legends.

Supporting information

Video 1

Video 2

Video 3

Video 4

Video 5

Video 6

Video 7

Video 8

Video 9

Video 10

Video 11

Video 12

Video 13

Video 14

Video 15

Video 16

Video 17

Video 18

Video 19

Video 20

Data and code availability

Owing to size restrictions, the experimental image data have not been uploaded to a repository but are available from the corresponding author on request. The codes generated during this study are available on GitHub (*https://github.com/agnikdasgupta/Sema7A_regulates_neural_circuitry*).

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Author Contributions

A.D. initiated the project, conducted experiments, analyzed data, and quantified results. C.C.R developed the Python codes for the network topology analysis and quantified those results, as well as analyzed published transcriptomics data. S.P.P. developed the Python codes for three-dimensional visual representation of sensory arbors and quantified arbor morphologies. L.M.S. conducted preliminary experiments. A.J. plotted Sema7A intensity measurements. A.D. wrote the paper with contributions from C.C.R and A.J.H.

Declaration of Interests

The authors declare no competing interests.

Video 1. Sema7A is enriched at the hair cell base and sensory-axon interface. A through-focus scan of a 4 dpf control neuromast depicts the hair cells (green) that are innervated by the sensory axons (magenta) at their basolateral surfaces where the Sema7A protein (orange) is highly enriched. Scale bar, 5 µm.

Videos 2 and 3. FM4-64 labels the hair cells of both control and sema7a-/- mutant neuromasts. Through-focus scans of 4 dpf control (Video 2) and *sema7a*$^{-/-}$ mutant (Video 3) neuromasts depict the hair cells (green) that have incorporated FM4-64. The sensory arbors (magenta) are tightly associated with the clustered hair cells in the control neuromast but show aberrant arborization in the *sema7a*$^{-/-}$ larvae.

Video 4. Three-dimensional arborization patterns in control and *sema7a*$^{-/-}$ neuromasts. Lateral views depict three-dimensionally rendered combined skeletonized network traces from 27 control and 27 *sema7a*$^{-/-}$ mutant neuromasts at 4 dpf.

Video 5. Sensory axonal projections contact an embryonic muscle progenitor cell expressing Sema7A^{sec}. A through-focus scan of a 3 dpf larva depicts a muscle progenitor cell in the dermomyotome that expresses the ectopic Sema7A^{sec} (orange) and the sensory axonal projection (magenta) that makes intimate contact with it. Scale bar, 20 µm.

Video 6. Volumetric surface reconstruction of a sensory arbor and an embryonic muscle progenitor expressing Sema7A^{sec}. Reconstruction of the muscle progenitor cell (orange) at 3 dpf reveals its intimate association with an attracted sensory arbor (magenta).

Video 7. Volumetric surface reconstruction of a sensory arbor and the mature myofibers expressing Sema7A^{sec}. Reconstruction of mature myofibers expressing ectopic Sema7A^{sec} (orange) at 3 dpf shows their close association with the attracted sensory arbors (magenta).

Video 8. Sensory axonal projections remain in proximity to myofibers expressing Sema7A^{sec}. A through-focus scan of a 3 dpf larva depicts mature myofibers expressing exogenous Sema7A^{sec} deep in the myotome (orange) and the sensory axonal projection (magenta) that remains nearby.

Video 9. Aberrant axonal projections reenter the lateral-line nerve while following an ectopic Sema7A^{sec} source. A through-focus scan of a 3 dpf larva depicts cells expressing exogenous Sema7A^{sec} (orange) that guides the aberrant sensory axonal projection (magenta) back into the lateral-line nerve. Scale bar, 20 µm.

Videos 10, 11, and 12. Ectopic Sema7A^{sec} expression induces aberrant neurite projections from the lateral-line nerve. Through-focus scans of 3 dpf larvae depict three individual incidents of ectopic *sema7a*^{sec} (orange) integrations that occurred distant from the neuromasts. In each case, myofibers express ectopic Sema7A^{sec} protein (orange) that attracts neurite extensions from the lateral-line nerve (magenta). Scale bars, 20 µm.

Videos 13 and 14. Volumetric reconstructions depict close contacts between the sensory neurites and the ectopic Sema7a^{sec} expressing cells. Volumetric reconstructions of 3 dpf larvae depict two individual incidents of ectopic *sema7a*^{sec} (orange) integrations that occurred distant from the neuromasts. In each case, myofibers or random mosaic of cells in the dermomyotome express ectopic Sema7A^{sec} protein (orange) that attracts neurite extensions from the lateral-line nerve (magenta).

Video 15 and 16. Time-lapse video microscopies depict the dynamics between the sensory neurites and the ectopic Sema7a^{sec} expressing cells. Time-lapse images of 3 dpf larvae depict two individual incidents where the ectopic Sema7A^{sec} sources (orange) induce directed branching of the sensory neurites (magenta) from the lateral-line nerve toward itself. Within each time-lapse videos, individual panel depicts the ectopic integration site and the associated sensory neurites from three different planes, XY, XZ, and YZ. Scale bars, 20 µm.

Videos 17, 18, and 19. Sema7A-GPI is incapable to impart sensory axon guidance from a distance. Through-focus scans of 3 dpf larvae depict three individual incidents of ectopic *sema7a-GPI* integrations that occurred near the sensory arbor network. In each case, myofibers robustly express ectopic Sema7A-GPI protein (orange) but the sensory arborization pattern (magenta) remain unperturbed. Scale bars, 20 µm.

Video 20. Postsynaptic assemblies are enriched in the loops and near the nodes of a sensory arbor network. Through-focus scans of a 4 dpf neuromast depict the sensory arbor network (magenta) where the postsynaptic assemblies (yellow) are primarily located in the loops of the arbor network and are enriched near the nodes.

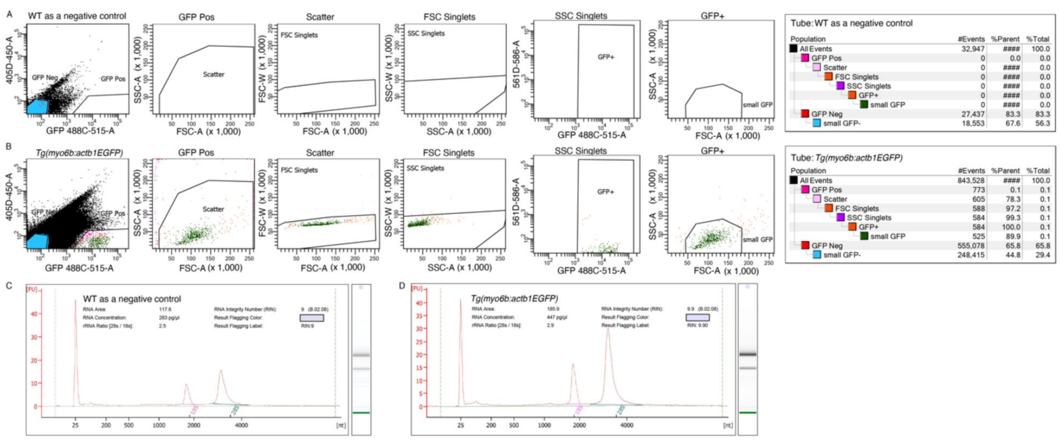

Figure 1—figure supplement 1.

Isolation of GFP-expressing hair cells.

(A,B) Representative results from florescence-activated cell sorting demonstrate the efficient discrimination of GFP^{+} (green) hair cells from *Tg(myo6b:actb1-EGFP)* larval tails. Around 400 tails were utilized to isolate hair cells. Wild type (WT) controls confirm the selectivity of the gates. (C,D) Plots represent the RNA integrity numbers (RINs) of the isolated total RNA from WT and transgenic larvae. High RIN values suggest minimal degradation of isolated RNAs from both samples.

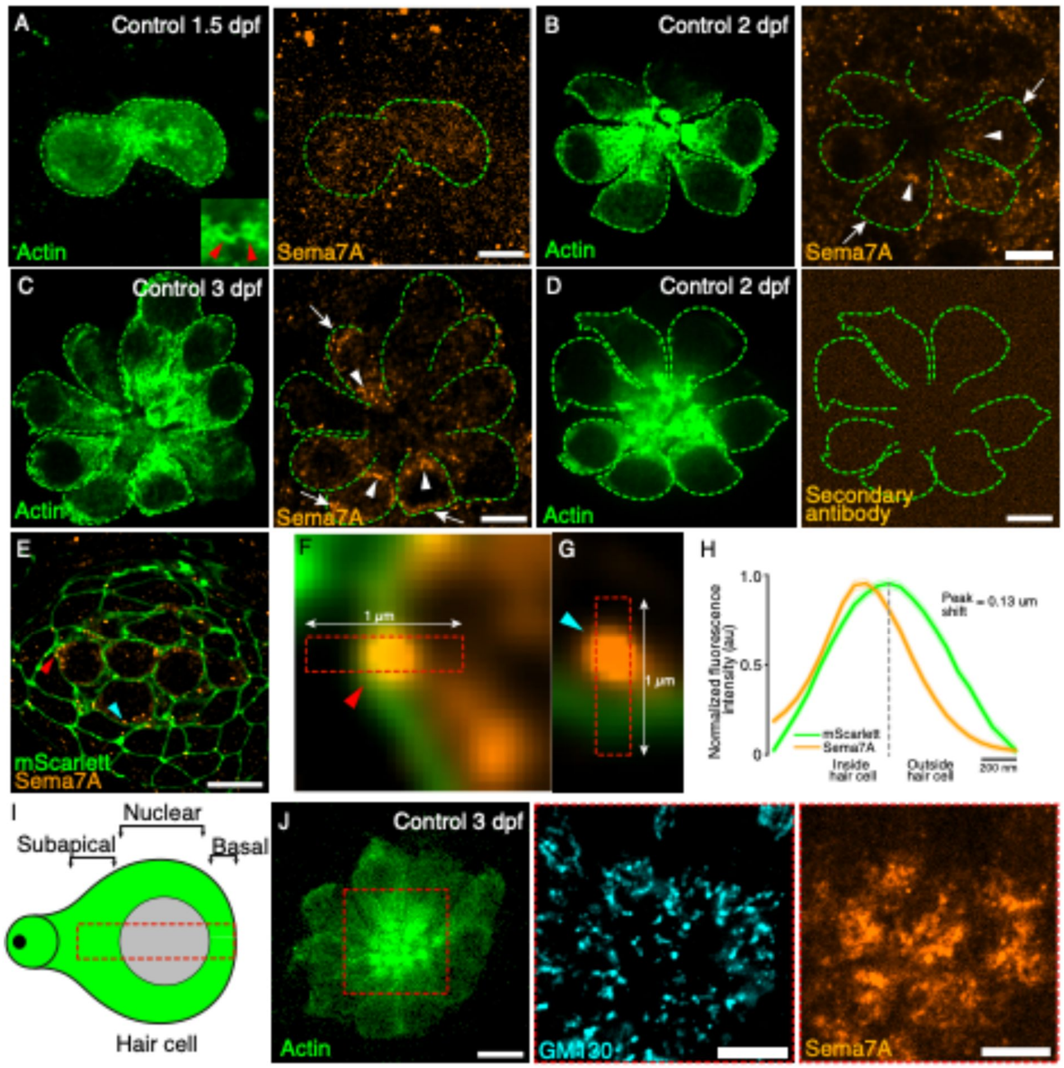

Figure 1—figure supplement 2.

Expression of Semaphorin7A in the lateral line neuromast.

(A-C) Surface micrographs show hair cells marked with actin-GFP (green dashed lines) and the localization of Sema7A protein (orange) in the subapical (arrowheads) and basal regions (arrows) of the corresponding hair cells from neuromasts of developing control larvae. The inset in panel A (red arrowheads) depicts the apices of the immature hair cell pair. (D) Immunolabeling with only the secondary antibody fails to detect any Sema7A signal. (E) A micrograph at a nuclear plane of a 4 dpf neuromast depicts membrane-tethered Scarlet (green) in all cells of the neuromast and the Sema7A protein (orange, arrowheads) in centrally clustered hair cells. (F) The Sema7A protein is located at the basal membrane (red arrowhead). (G) The Sema7A protein is also observed adjacent to the basal membrane, potentially in a vesicle (cyan arrowhead). (H) A plot quantifies the distribution of average Sema7A intensity across the hair cell's basal membrane. The results stem from 100 hair-cell membranes of 10 neuromasts from 4 dpf larvae. (I) A schematic diagram of a single hair cell depicts the three distinct regions along the apicobasal axis of the cell. Sema7A intensity was measured along this apicobasal axis (red dashed line). (J). A surface micrograph depicts hair cells (green, left) whose subapical region (red dashed line) harbors the Golgi network (cyan, middle) where the Sema7A protein is enriched (orange, right). Scale bars, 5 µm; au, arbitrary unit; means ± SEMs.

926

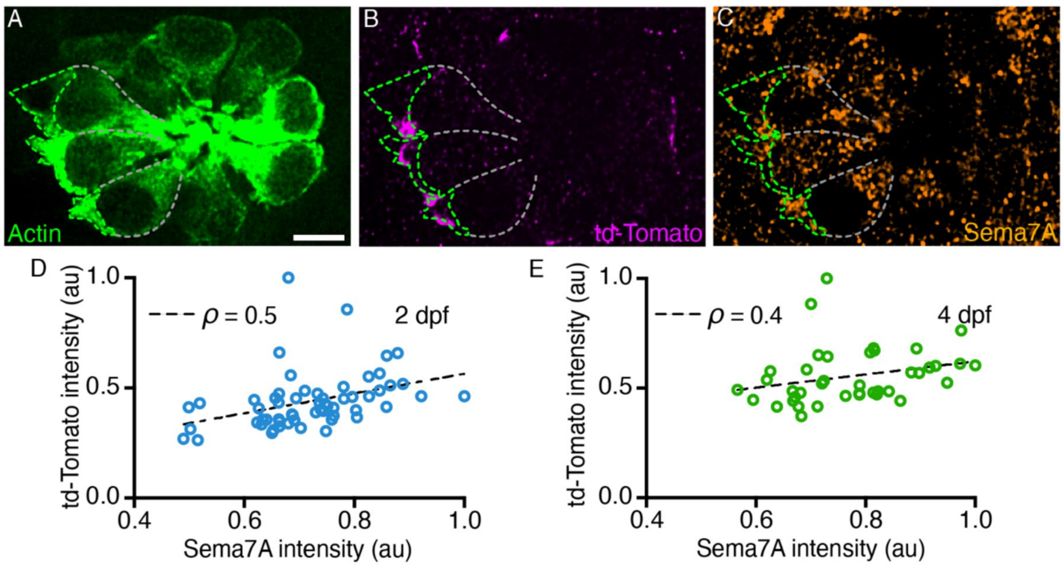

Figure 1—figure supplement 3.

Sema7A progressively enriches at the hair cell base with the sensory axons.

(A) A micrograph at a nuclear plane of a 3 dpf neuromast depicts several actin-GFP (green) labeled hair cells. The lateral (white dashed line) and the basal (green dashed line) surfaces of three hair cells are outlined. (B) The td-Tomato$^+$ sensory arbors contact the basal regions (green dashed line) of three hair cells. (C) The Sema7A is enriched in the basal regions (green dashed line) of the hair cells that are in contact with the sensory arbors. (D,E) At the base of the hair cell, intensities of the td-Tomato$^+$ sensory arbors positively correlate with the Sema7A intensities. The results stem from 66 hair cells of 15 neuromasts from 2 dpf larvae and 40 hair cells of 11 neuromasts from 4 dpf larvae. Scale bars, 5 μm; au, arbitrary unit; ρ, Spearman's correlation coefficient.

```
Mus              MTPPPPGRAAPSAPRARVLSLPARFGLPLRLRLLLVFWVAAASAQGHSRSGPRISAVWKG
Danio            ------------------MIRHYSSLKFKGNLAYPWLIYGACFQCVFCSCNPRLTLKQD
                                    :    .* :: .*   : : .*. *    *     :: :.

Mus              QDHVDFSQPEPHTVLFHEPGSFSVWVGGRGKVYHFNFPEGKNASVRTVNIGSTKGSCQDK
Danio            NAPLNYNTNYSHTVLFHEEGSKSLFVGGSNIVLHFDVDSYEILENFTLNANLPR---CGE
                 :  :::.   .******* ** *::*** . * **:. . :  .  *:* . .:    .:

Mus              QDCGNYITLLERRGNGLLVCGTNARKPSCWNLVN---DSVMMSLGEMKGYAPFSPDENSL
Danio            SSCENVVTVIETFQNCKFVCGTNGEEPQCWELYPRESNHSVVKSVEGTGISPHSSEQNSL
                 ..* * :*::*   *  :*****..:*.**:*     :  ::.  * .* :*.*.::***

Mus              VLFEGDEVYSTIRKQEYNGKIPRFRRIRGESELYTSDTVMQNPQFIKATIVH-QDQAYDD
Danio            SLVADGDLYAAAPLYSDGTLLQFRRKAGKRINVWMHDQWVSEPTFISSFLAKRKNDSLNE
                  *. ..::*::    . .  :    *:    . :::  *  :.:* **.: :.: :::: ::

Mus              KIYYFFREDNPDKNPEAPLNVSRVAQLCRGDQGGESSLSVSKWNTFLKAMLVCSDAATNR
Danio            KIYVLFREKNSDTSPEADPWISRVARVCKIDEGGSKQFLQNIWTSFLKARLVCGIPKESL
                 *** :***.*.*..***   :****::*: *:**...:   . *.:**** ***. .   .

Mus              NFNRLQDVFLLPDPSGQWRDTRVYGVFSNPWNYSAVCVYSLGDIDRVFRTSSLKGYHMGL
Danio            YFNRLQDVFVQH--AEDWRDSLVYALFSSSWNSTAVCIYSLADLDDIFEKSNYKGFSEAI
                  ********:    : :***: **.:**..** :***:***.*:* :*..*. **:  .:

Mus              PNPRPGMCLPKKQPIPTETFQVADSHPEVAQRVEPMGPLKTPLFHSKYHYQKVVVHRMQA
Danio            PNPRPGKCVENSNSLPIPTLKIIKDYPEMKDWIHPIQ-KDTPFFTSSKNFTKIVVDQIRA
                 ****** *: :.:.:*  *::: ..:**: : :.*:   .**:* *. :: *:**.:::*

Mus              SNGETFHVLYLTTDRGTIHKVVESGDQDHSFVFNIMEIQPFHRAAAIQAISLDADRRKLY
Danio            LDENVYNVLLLATDDGMIFKILQHG----SKPFIISETHLCNGCAPIQSMKLDSKKRKLF
                  : :.::** *:** * *.*::: *     *  * * * :  : .*.**::.**:.:***:

Mus              VTSQWEVSQVPLDMCEVYSGGCHGCLMSRDPYCGWDQDRCVSIYSS--QRSVLQSINPAE
Danio            VGYPGQMSVLELQRCQDYGASCEECVLSRDPYCAWSEKGCTSQTRGGIQNISDGNVTKVC
                 *    ::* : *: *: *...*. *::******.*.:. *.*   .  *.    .:. .

Mus              PHRECPNPKPDE------APLQKVSLARNSRYYLTCPMESRHATYLWRHEENVEQSCEPG
Danio            DHKSVGRVRRDAQQSSTAPSRIQHTVATDVPFYLSCPVDSHHASYRWEHGG---KSNPCQ
                  *:.  . : *       ..  : ::* :  :**:**::*:**:* *.*     :*

Mus              HQSPSCILFIENLTARQYGHYRCEAQEGSYLREAQHWELLPE-DRALAEQLMGHARALAA
Danio            QTQSEYLLLIPAMTAENYGNYSCVSQERDYIKVVREYELLAKSVFNDAFKLKIQDRQLMV
                 : ... :*:*  :**.:**:* * :** :** .*:: .:.:***.:     * :*  : * * .

Mus              SFWLGVLPTLILGLLVH
Danio            AFVTSAFYLCSL-----
                 :*  ..:    *
```

Sequence 1: Mus musculus   664 aa

Sequence 2: Danio rerio   640 aa

Immunogen: Mouse myeloma cell line NS0-derived recombinant mouse Semaphorin 7A Glutamine 45-Alanine 646, which corresponds to the Glutamine 26-Methionine 627 of zebrafish Semaphorin 7A. We predict that the antibody recognizes both variants of the *sema7a* encoded proteins. The Sema domains in both mouse and zebrafish are highlighted in orange.

Figure 1—figure supplement 4.

Alignment of the mouse and the zebrafish Sema7A protein sequences generated by Clustal Omega.

(A) Multiple sequence alignment depicts the conserved Sema domain (orange) and the immunogenic region for the Sema7A antibody in mouse, and its equivalent region in zebrafish.

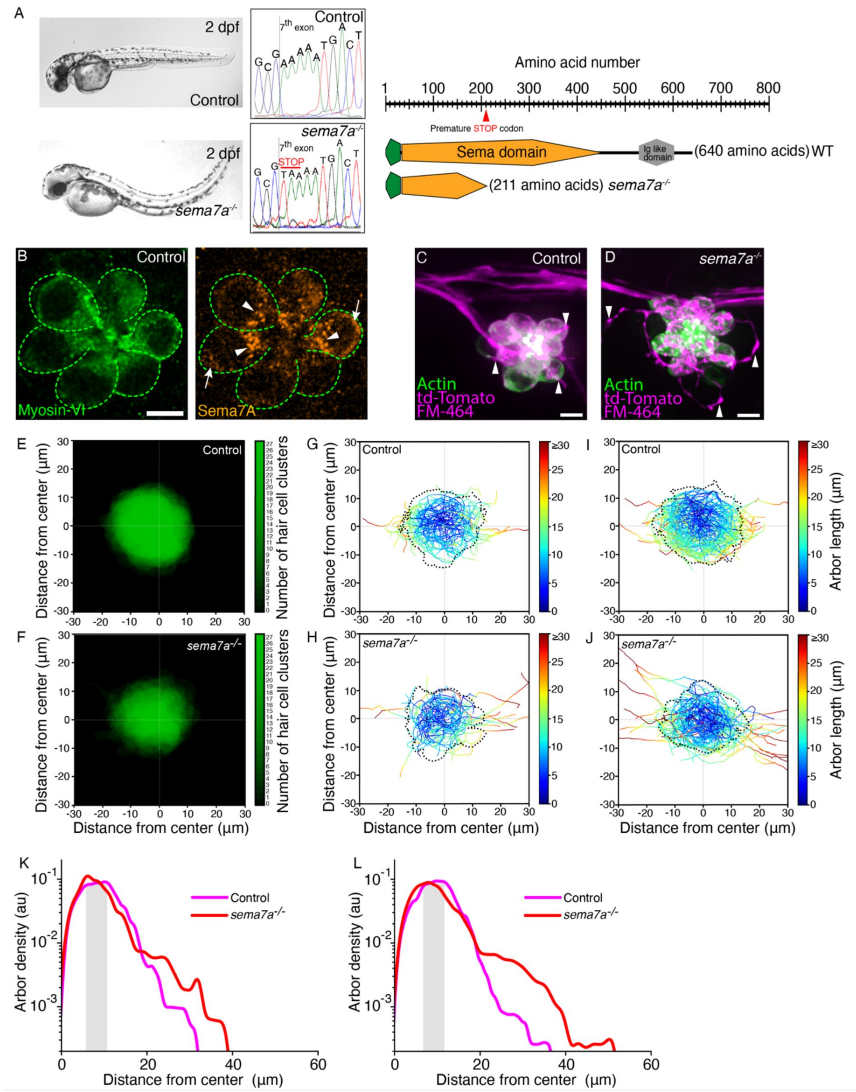

Figure 2—figure supplement 1.

sema7a$^{-/-}$ mutants exhibit aberrant sensory-axon arborization throughout development.

(A) Micrographs depict the overall morphologies and *sema7a* genomic profiles of 2 dpf control (top) and *sema7a*$^{-/-}$larvae (bottom). The schematic diagram on the right depicts the mutation site and the resulting early truncation of the encoded protein. (B) In a surface micrograph of a 3 dpf control neuromast, myosin VI (green) marks hair cells (green dashed lines). Sema7A (orange) protein is highly enriched in the subapical region (arrowheads) and basolateral (arrows) region of the hair cells. (C,D) Micrographs depict maximum intensity projections of 4 dpf neuromasts from control and *sema7a*$^{-/-}$larvae where the hair cells (green) are stained with FM4-64 (magenta). The sensory arbors (arrowheads, magenta) tightly enwrap the clustered hair cells in the control neuromast but show aberrant arborization in the *sema7a*$^{-/-}$ larvae. (E,F) Micrographs depict combined 4 dpf hair cell clusters, each of whose centers arise located at (0,0). The X- and Y-coordinates represent the anteroposterior (AP) and dorsoventral (DV) axes of the larva, respectively. Positive values of the X- and Y-coordinates represent posterior and ventral directions, respectively. Combined hair cell clusters from 27 control and 27 *sema7a*$^{-/-}$ mutant neuromasts are represented. (G,H) Micrographs depict the skeletonized networks of the combined 2 dpf hair cell clusters from 17 control and 17 *sema7a*$^{-/-}$ mutant neuromasts. (I,J) Micrographs depict the skeletonized networks of the combined 3 dpf hair cell clusters from 29 control and 29 *sema7a*$^{-/-}$ mutant neuromasts. (K) A plot quantitates the densities of the sensory arbors around the center of the combined hair cell clusters for 33 control (magenta) and 17 *sema7a*$^{-/-}$ (red) neuromasts at 2 dpf. The shaded area marks the region proximal to the boundary of the combined hair cell cluster. (L) A plot quantitates the densities of the sensory arbors around the center of the combined hair cell clusters for 29 control (magenta) and 53 *sema7a*$^{-/-}$ (red) neuromasts at 3 dpf. The shaded area marks the region proximal to the boundary of the combined hair cell cluster. Scale bars, 5 μm; au, arbitrary units.

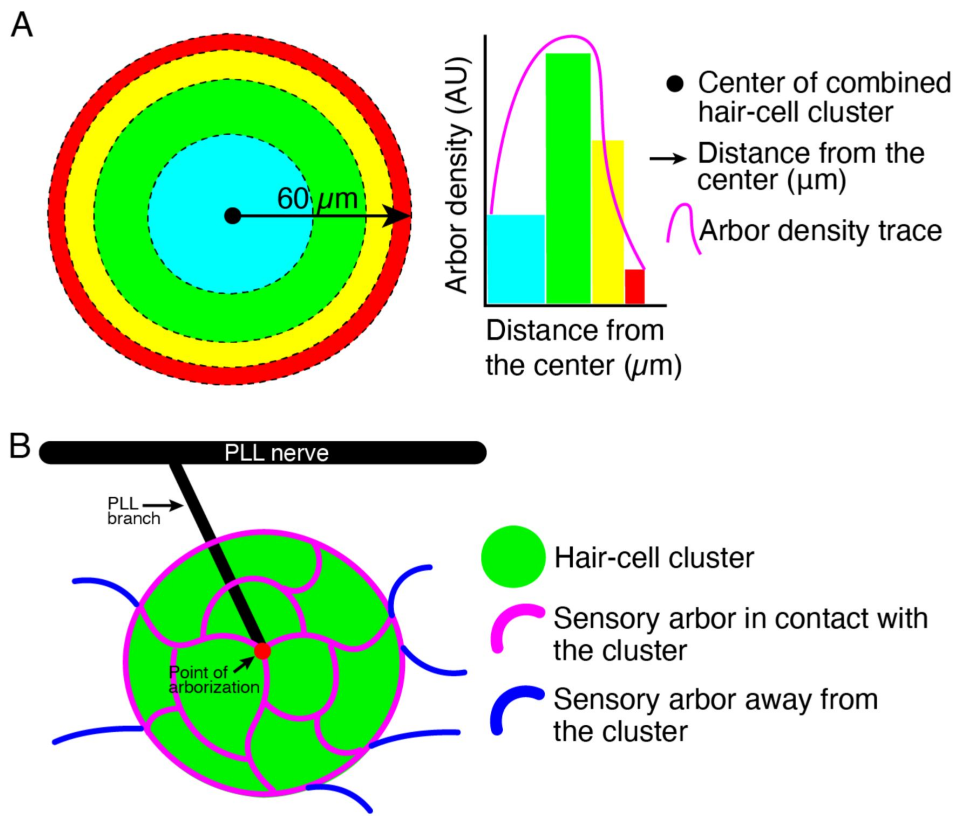

Figure 2—figure supplement 2.

Quantification of sensory arbor distributions and contacts with hair cell clusters.

(A) A schematic drawing of the combined hair cell cluster and surrounding region depicts concentric sections of equal area in distinct colors (cyan, green, yellow, and red). The arbor density is plotted as a histogram against the distance (black arrow) from the center (black circle) of the combined hair cell cluster. The histogram is represented as a density trace graph (magenta). (B) A schematic drawing of a posterior lateral-line neuromast depicts the posterior lateral-line (PLL) nerve (black), posterior lateral-line branch (black), the point or arborization, the sensory arbors (magenta) that contact the hair cell cluster (green), and the sensory arbors that project away (blue) from the hair cell cluster.

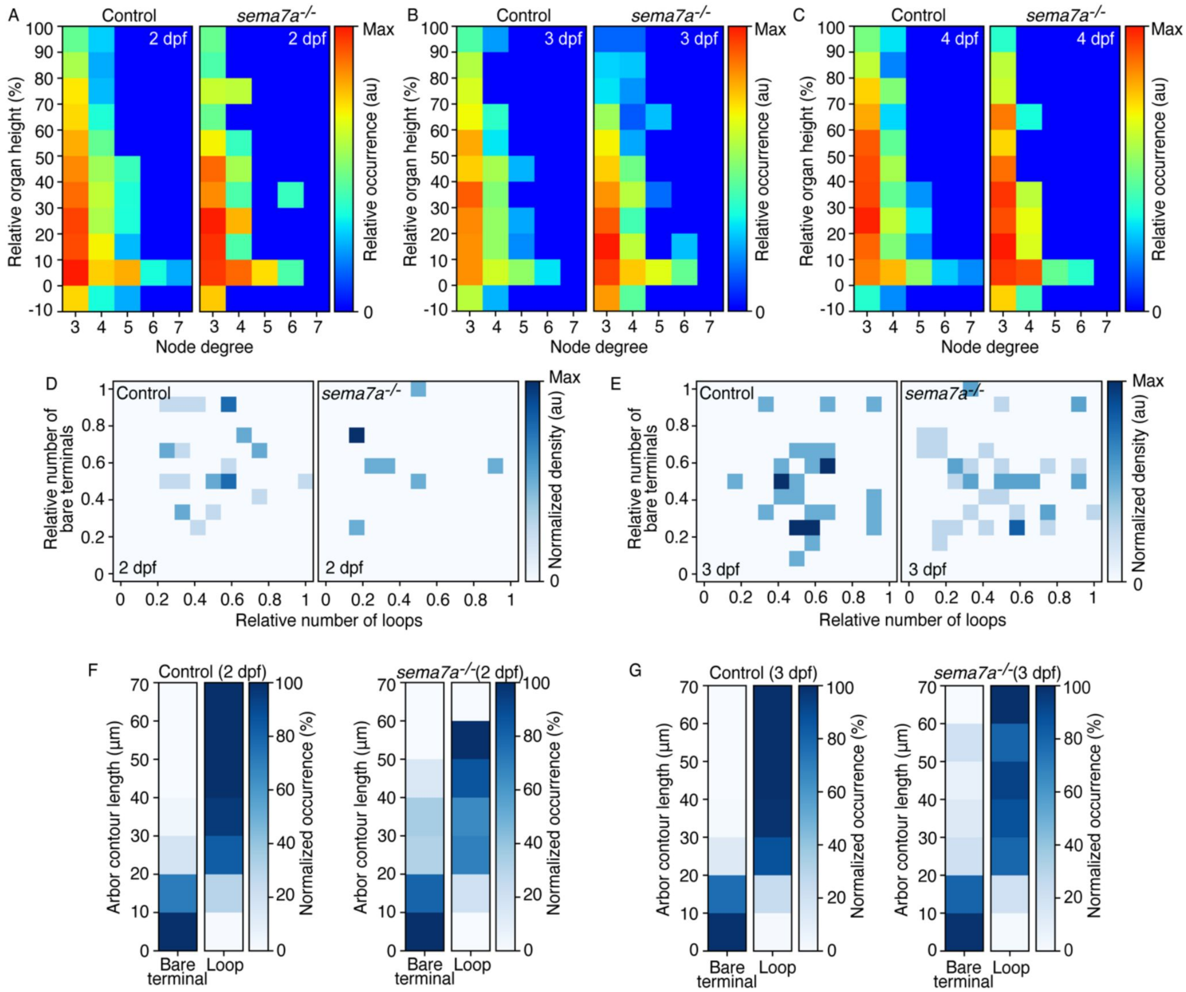

Figure 3—figure supplement 1.

Quantification of topological features across developmental stages and genotypes.

(A-C) The distribution of node degrees as a function of the height in hair cell clusters shows the difference in the network's hierarchical organization between control and *sema7a*$^{-/-}$ larvae. Nodes of low degree are commonly observed between genotypes across developmental stages (A, B), but their distribution along the cluster height became significantly different at 4 dpf (C), with nodes in control larvae localizing higher along the axis than in the mutants. In this and the following two panels, 33, 29, and 35 neuromasts were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively and 17, 53, and 27 neuromasts were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$mutant larvae, respectively (D, E) The correlations between the relative number of bare terminals and loops from control and *sema7a*$^{-/-}$ neuromasts reveal dispersive distributions at both 2 dpf (D) and 3 dpf (E) stages, suggesting that the networks are undergoing reorganization and yet to adopt an ordered topology. (F,G) In the distribution of arbors between bare terminals and loops with respect to their contour lengths, mutants' arbors display significantly higher propensity to remain as bare terminals compared to the controls at both 2 dpf (F) and 3 dpf (G). The samples were identical to those in the previous panel. Statistical analyses are displayed in **Table 1**.

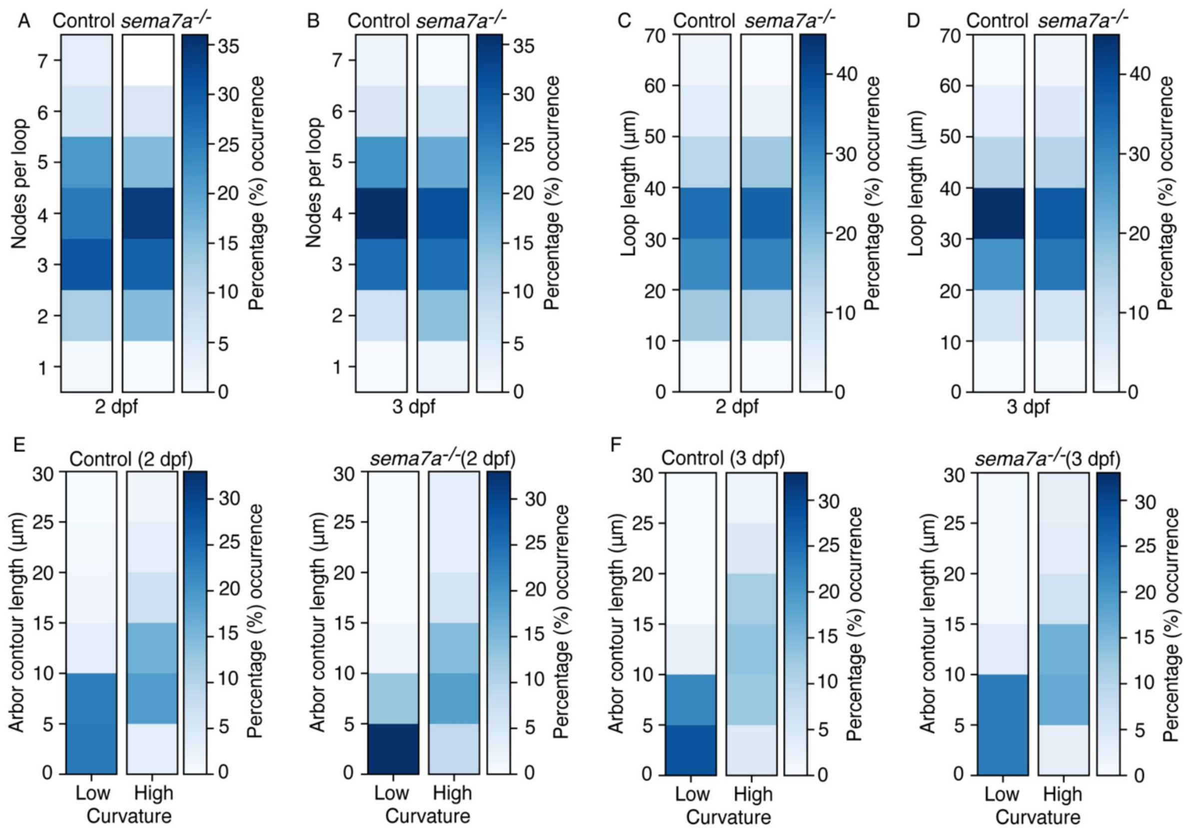

Figure 3—figure supplement 2.

Quantification of topological features between control and *sema7a*$^{-/-}$ larvae across developmental stages.

(A, B) Quantifications of the number of nodes per loop demonstrate a broader, but not significantly different, distribution in the mutants than the controls at both 2 dpf (A) and 3 dpf (B). In this and the following panels, 33 and 29 neuromasts were analyzed from 2 dpf and 3 dpf control larvae, respectively and 17 and 53 neuromasts were analyzed from 2 dpf and 3 dpf *sema7a*$^{-/-}$ mutant larvae, respectively. (C, D) Quantifications of loop lengths from the same larvae indicate a wider, but not significantly different, distribution in the mutants than the controls at both 2 dpf (C) and 3 dpf (D). (E, F) The distributions of bare terminals between low-and high-curvature groups from the same larvae show higher occurrences of long, curved arbors in the control and short, linear arbors in the mutants at both 2 dpf (E) and 3 dpf (F) stages. Statistical analyses are displayed in **Table 1**.

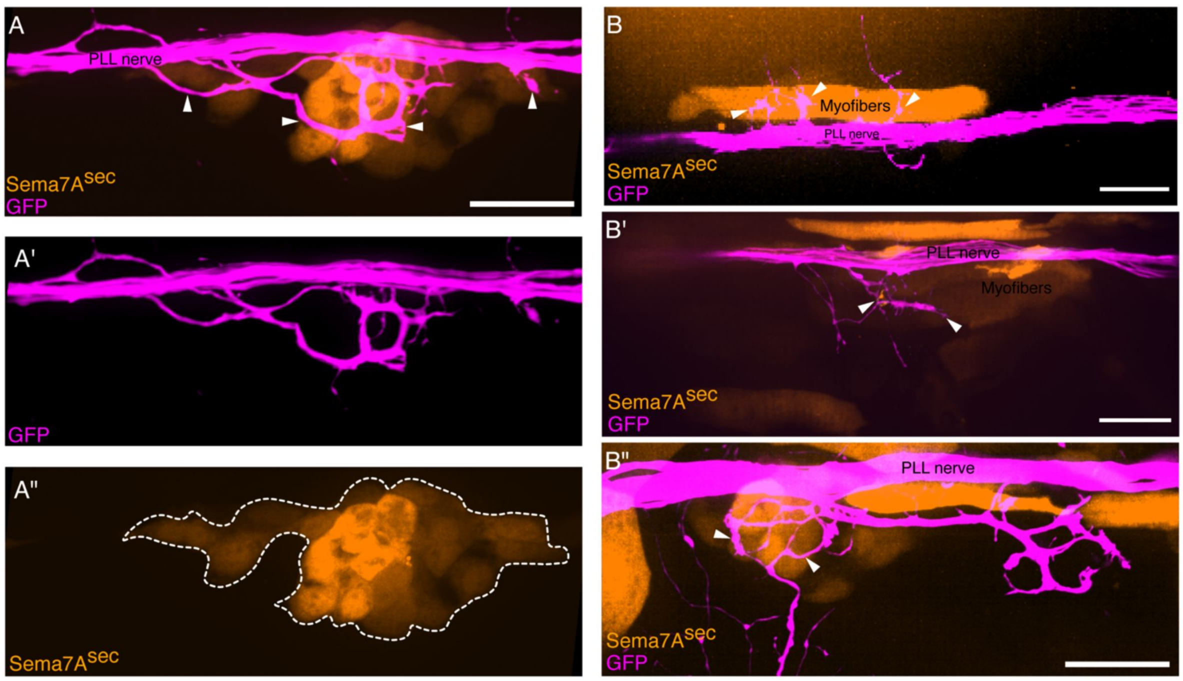

Figure 4—figure supplement 1.

Ectopic Sema7A^{sec} diffusive cues induce aberrant sensory neurite extension.

(A-A”) In a micrograph of an ectopically expressing Sema7A^{sec} (orange) larva, the sensory arbors (magenta) extend multiple axonal projections (arrowheads) that reenter the lateral-line nerve while following an ectopic Sema7A source. Maximal-intensity projections of the sensory arbors and the ectopically expressing Sema7A cells (white dashed line) are shown in A’ and A”, respectively. (B-B”) Micrographs depict three individual incidents of ectopic *sema7a*^{sec} (orange) integrations that occurred distant from neuromasts. In each case, a set of myofibers expresses the ectopic Sema7A^{sec} protein that induces neurite extensions (arrowheads) from the lateral-line nerve (magenta) toward itself. Scale bars, 20 µm.

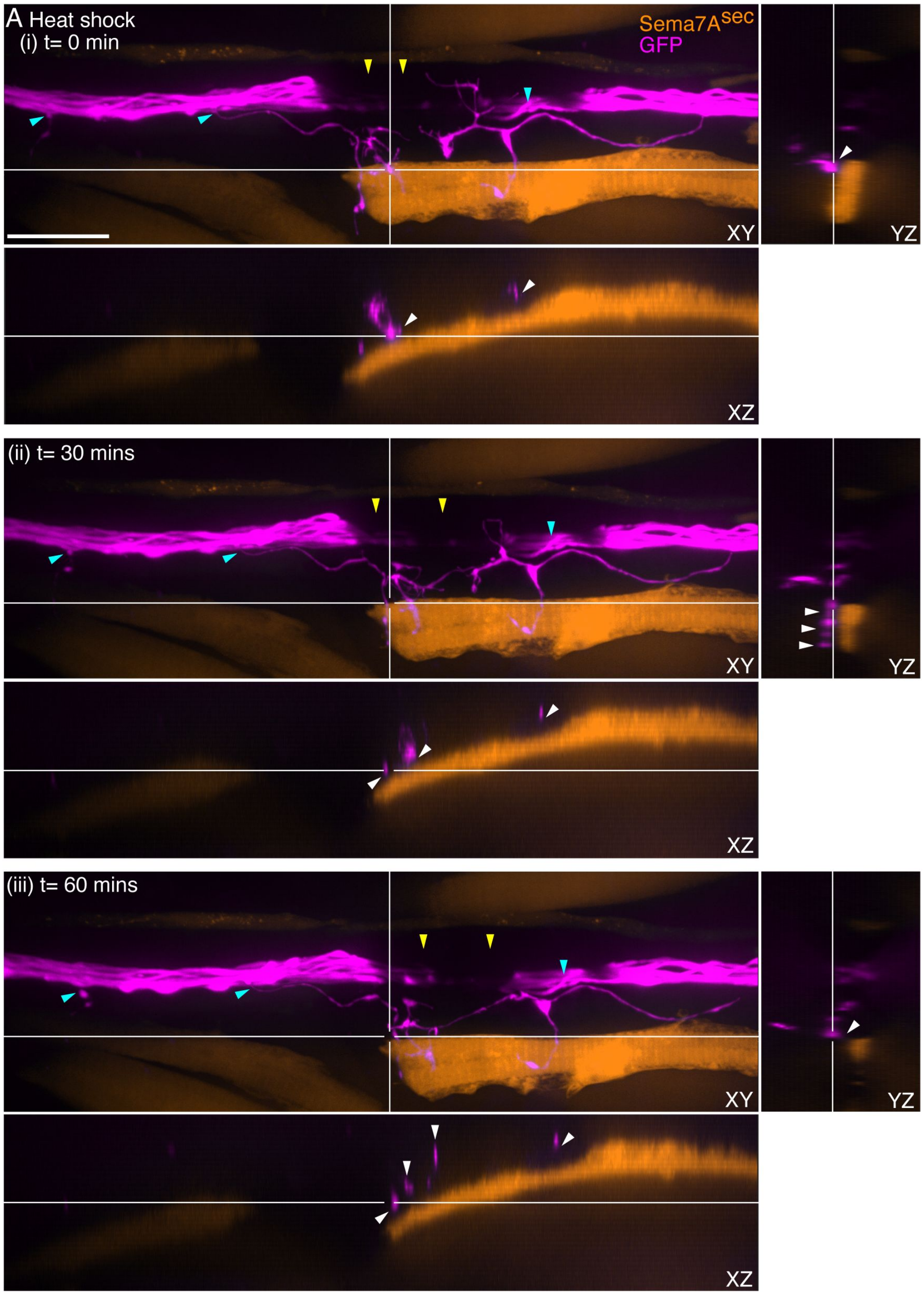

Figure 4—figure supplement 2.

Ectopic integration of Sema7A^{sec} distant from the neuromast induces sensory neurite extension.

(A) Micrographs of a section of the lateral-line nerve between two neuromasts from a 3-dpf heat-shocked larva depicts a few extended neurites (magenta) reaching toward the cells expressing the ectopic Sema7A^{sec} (orange). The Sema7A^{sec} integration site with the sensory axonal projections was imaged for one hour to visualize axonal dynamics. Micrographs (i-iii) at three distinct time-points from the time-lapse video microscopy show directed branching (cyan arrowheads) of the sensory neurites from the lateral-line nerve toward the ectopic Sema7A^{sec} sources. Each panel depicts the ectopic integration site and the associated sensory neurites from three different planes, XY, XZ, and YZ. White arrowheads in XZ and YZ planes highlight the close association of the sensory axons with the ectopic Sema7A^{sec} sources. Melanocytes (yellow arrowheads) along the horizontal myoseptum intermittently block the visibility of the lateral-line nerve. Scale bar, 20 μm.

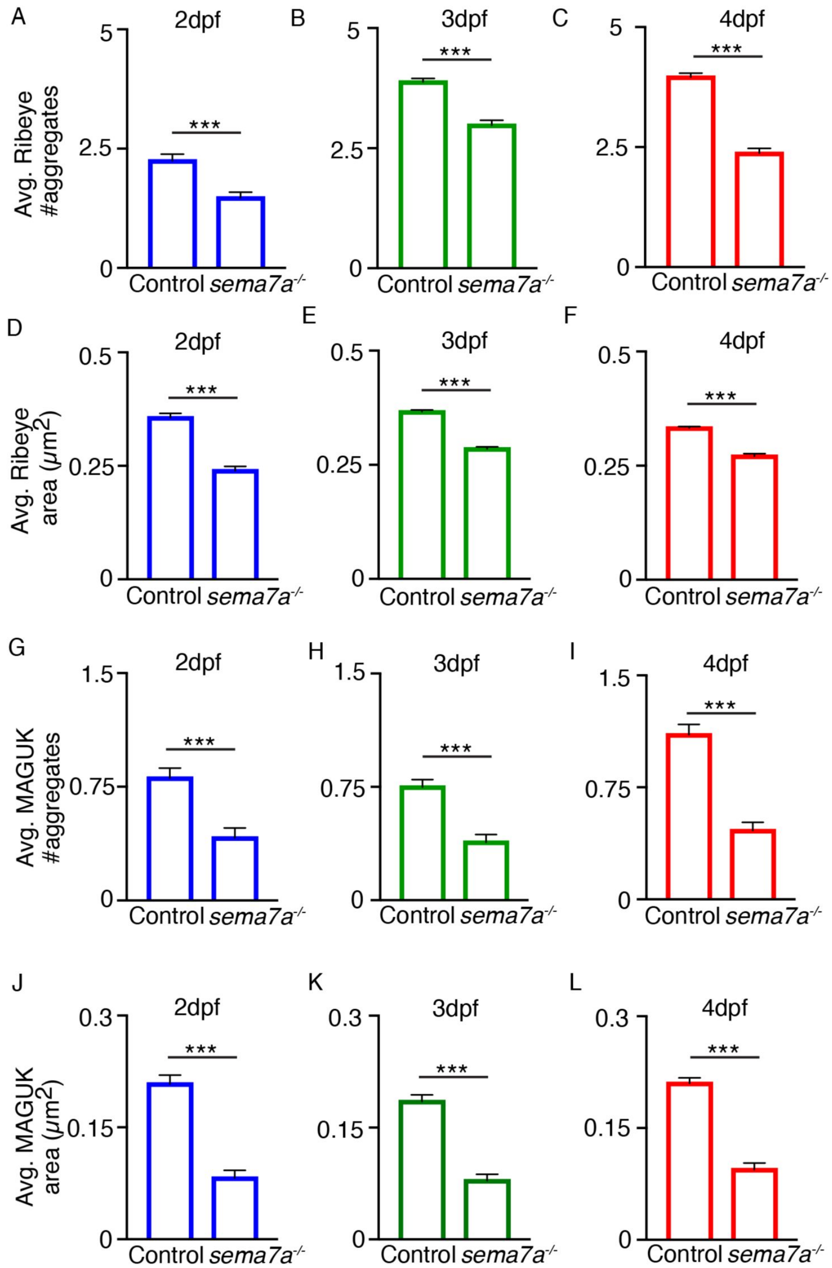

Figure 6—figure supplement 1.

sema7a$^{-/-}$ mutants display impaired synaptic assembly.

(A-F) The plots quantitate the average numbers and areas of presynaptic aggregates from control and *sema7a*$^{-/-}$ neuromasts. Significant decreases in the numbers and areas of presynaptic aggregates are observed across development. Three hundred and forty-one, 1229, and 1286 presynaptic aggregates were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively. Two hundred and thirty-five, 643, and 419 presynaptic aggregates were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$ mutant larvae, respectively. (G-L) The plots quantitate the average numbers and areas of postsynaptic aggregates from control and *sema7a*$^{-/-}$ neuromasts. Significant decreases in the numbers and areas of postsynaptic aggregates occurred during development. Three hundred and ninety-seven, 1243, and 1300 postsynaptic aggregates were analyzed from 2 dpf, 3 dpf, and 4 dpf control larvae, respectively. Three hundred and four, 651, and 451 postsynaptic aggregates were analyzed from 2 dpf, 3 dpf, and 4 dpf *sema7a*$^{-/-}$ mutant larvae, respectively. Means ± SEMs; *** implies $p < 0.001$.

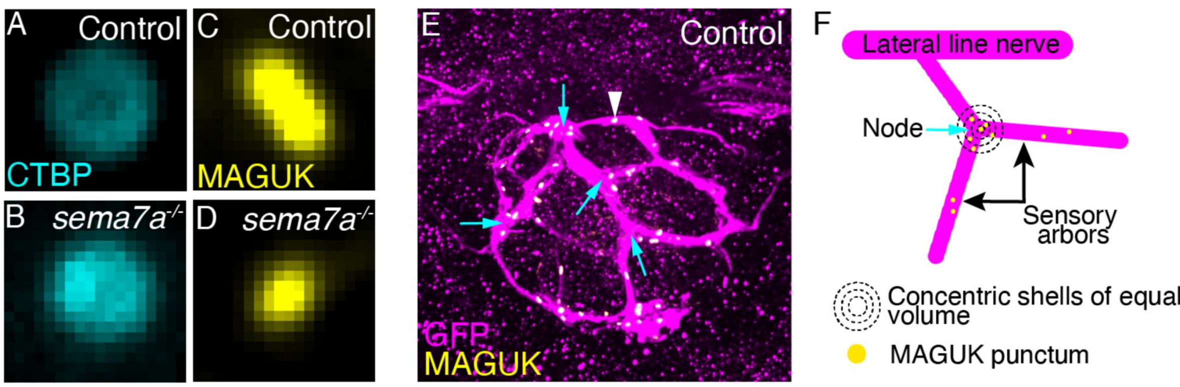

Figure 6—figure supplement 2.

Morphologies of presynaptic and postsynaptic assemblies and the distribution of postsynaptic aggregates across the sensory arbor network.

(A,B) The micrographs depict single presynaptic aggregate marked by CTBP in control (A) and in sema7a-/-mutant (B) larvae at 4 dpf. (C,D) The micrographs depict single postsynaptic aggregate marked by MAGUK in control (C) and in sema7a-/-mutant (D) larvae at 4 dpf. (E) A maximal-intensity projection micrograph depicts the sensory arbor network (magenta) and the enrichment of the postsynaptic assemblies (yellow, white arrowhead) near the nodes (cyan arrows). (F) A schematic diagram of the sensory arbor network depicts the enrichment of the MAGUK punctae near the node (cyan arrow). The three-dimensional space surrounding each node was divided into concentric shells of equal volume to measure the distribution of MAGUK-punctae density across the network.

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Reviewer #1 (Public Review):

Dasguta et al. have dissected the role of Sema7a in fine tuning of a sensory microcircuit in the posterior lateral line organ of zebrafish. They attempt to also outline the different roles of a secreted verses membrane-bound form of Sema7a in this process. Using genetic perturbations and axonal network analysis, the authors show that loss of both Sema7a isoforms causes abnormal axon terminal structure with more bare terminals and fewer loops in contact with presynaptic sensory hair cells. Further, they show that loss of Sema7a causes decreased number and size of both the pre- and post-synapse. Finally, they show that overexpression of the secreted form of Sema7a specifically can elicit axon terminal outgrowth to an ectopic Sema7a expressing cell. Together, the analysis of Sema7a loss of function and overexpression on axon arbor structure is fairly thorough and revealed a novel role for Sema7a in axon terminal structure. However, the connection between different isoforms of Sema7a and the axon arborization needs to be substantiated. Furthermore, the effect of loss of Sema7a on the presynaptic cell is not ruled out as a contributing factor to the synaptic and axon structure phenotypes. These issues weaken the claims made by the authors including the statement that they have identified dual roles for the GPI-anchored verses secreted forms of Sema7a on synapse formation and as a chemoattractant for axon arborization respectively.

https://doi.org/10.7554/eLife.89926.2.sa3

Reviewer #2 (Public Review):

In this work, Dasgupta et al. investigates the role of Sema7a in the formation of peripheral sensory circuit in the lateral line system of zebrafish. They show that Sema7a protein is present during neuromast maturation and localized, in part, to the base of hair cells (HCs). This would be consistent with pre-synaptic Sema7a mediating formation and/or stabilization of the synapse. They use sema7a loss-of-function strain to show that lateral line sensory terminals display abnormal arborization. They provide highly quantitative analysis of the lateral line terminal arborization to show that a number of specific topological parameters are affected in mutants. Next, they ectopically express a secreted form of Sema7a to show that lateral line terminals can be ectopically attracted to the source. Finally, they also demonstrate that the synaptic assembly is impaired in the sema7a mutant. Overall, the data are of high quality and properly controlled. The availability of Sema7a antibody is a big plus, as it allows to address the endogenous protein localization as well to show the signal absence in the sema7a mutant. The quantification of the arbor topology should be useful to people in the field who are looking at the lateral line as well as other axonal terminals. I think some results are overinterpreted though. The authors state: "Our findings demonstrate that Sema7A functions both as a juxtracrine and as a secreted cue to pattern neural circuitry during sensory organ development." However, they have not actually demonstrated which isoform functions in HCs (also see comments below). In addition, they have to be careful in interpreting their topology analysis, as they cannot separate individual axons. Thus, such analysis can generate artifacts. They can perform additional experiments to address these issues or adjust their interpretations.

https://doi.org/10.7554/eLife.89926.2.sa2

Reviewer #3 (Public Review):

The data reported here demonstrate that Sema7a defines the local behavior of growing axons in the developing zebrafish lateral line. The analysis is sophisticated and convincingly demonstrates effects on axon growth and synapse architecture. Collectively, the findings point to the idea that the diffusible form of sema7a may influence how axons grow within the neuromast and that the GPI-linked form of sema7a may subsequently impact how synapses form, though additional work is needed to strongly link each form to its' proposed effect on circuit assembly.

Comments on revised submission:

The revised manuscript is significantly improved. The authors comprehensively and appropriately addressed most of the reviewers' concerns. In particular, they added evidence that hair cells express both Sema7A isoforms, showed that membrane bound Sema7A does not have long range effects on guidance, demonstrated how axons behave close to ectopic Sema7A, and analyzed other features of the hair cells that revealed no strong phenotypes. The authors also softened the language in many, but not all places. Overall, I am satisfied with the study as a whole.

https://doi.org/10.7554/eLife.89926.2.sa1

Reviewer #4 (Public Review):

This study provides direct evidence showing that Sema7a plays a role in the axon growth during the formation of peripheral sensory circuits in the lateral-line system of zebrafish. This is a valuable finding because the molecules for axon growth in hair-cell sensory systems are not well understood. The majority of the experimental evidence is convincing, and the analysis is rigorous. The evidence supporting Sema7a's juxtracrine vs. secreted role and involvement in synapse formation in hair cells is less conclusive. The study will be of interest to cell, molecular and developmental biologists, and sensory neuroscientists.

https://doi.org/10.7554/eLife.89926.2.sa0

Author Response

The following is the authors' response to the original reviews.

> ***eLife assessment***
>
> *This study presents valuable findings on the roles of the axon growth regulator Sema7a in the formation of peripheral sensory circuits in the lateral line system of zebrafish. The*

> *evidence supporting the claims of the authors is solid, although further work directly testing the roles of different sema7a isoforms would strengthen the analysis. The work will be of interest to developmental neuroscientists studying circuit formation.*
>
> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *In this work, Dasguta et al. have dissected the role of Sema7a in fine tuning of a sensory microcircuit in the posterior lateral line organ of zebrafish. They attempt to also outline the different roles of a secreted verses membrane-bound form of Sema7a in this process. Using genetic perturbations and axonal network analysis, the authors show that loss of both Sema7a isoforms causes abnormal axon terminal structure with more bare terminals and fewer loops in contact with presynaptic sensory hair cells. Further, they show that loss of Sema7a causes decreased number and size of both the pre- and post-synapse. Finally, they show that overexpression of the secreted form of Sema7a specifically can elicit axon terminal outgrowth to an ectopic Sema7a expressing cell. Together, the analysis of Sema7a loss of function and overexpression on axon arbor structure is fairly thorough and revealed a novel role for Sema7a in axon terminal structure. However, the connection between different isoforms of Sema7a and the axon arborization needs to be substantiated. Furthermore, an autocrine role for Sema7a on the presynaptic cell is not ruled out as a contributing factor to the synaptic and axon structure phenotypes.*
>
> *Finally, critical controls are absent from the overexpression paradigm.*

Comments: Thank you for your valuable comments. We have analyzed the hair cell scRNA transcriptome data of zebrafish neuromasts from published works and have not identified known expression of receptors of the Sema7A protein, particularly PlexinC1 and Integrin β1 molecules (reference 4 and 15) in hair cells. This result suggests that the Sema7A protein molecule, either secreted or membrane-bound, does not possess its cognate receptor to elicit an autocrine function on the hair cells. Moreover, the GPI-anchored Sema7A lacks a cytosolic domain. So it is unlikely that Sema7A signaling directly induces the formation of presynaptic ribbons. We propose that the decrease in average number and area of synaptic aggregates likely reflects decreased stability of the synaptic structures owing to lack of contact between the sensory axons and the hair cells, which has been identified in zebrafish neuromasts (reference 38).

Thank you for pointing missing critical control experiments. Additional control experiments (lines 333-346) with a new figure (Figure 5) have been added.

> *These issues weaken the claims made by the authors including the statement that they have identified differential roles for the GPI-anchored verses secreted forms of Sema7a on synapse formation and as a chemoattractant for axon arborization respectively.*

Comments: We have rephrased our statement and argue in lines 428-430 that our experiments “suggest a potential mechanism for hair cell innervation in which a local Sema7Asec diffusive cue likely consolidates the sensory arbors at the hair cell cluster and the membrane-anchored Sema7A-GPI molecule guides microcircuit topology and synapse assembly.”

> *The manuscript itself would benefit from the inclusion of details in the text to help the reader interpret the figures, tools, data, and analysis.*

Comments: We have made significant revisions to the text and figures to improve clarity and consistency of the manuscript.

> ***Reviewer #2 (Public Review):***
>
> *In this work, Dasgupta et al. investigates the role of Sema7a in the formation of peripheral sensory circuit in the lateral line system of zebrafish. They show that Sema7a protein is present during neuromast maturation and localized, in part, to the base of hair cells (HCs). This would be consistent with pre-synaptic Sema7a mediating formation and/or stabilization of the synapse. They use sema7a loss-of-function strain to show that lateral line sensory terminals display abnormal arborization. They provide highly quantitative analysis of the lateral line terminal arborization to show that a number of specific topological parameters are affected in mutants. Next, they ectopically express a secreted form of Sema7a to show that lateral line terminals can be ectopically attracted to the source. Finally, they also demonstrate that the synaptic assembly is impaired in the sema7a mutant. Overall, the data are of high quality and properly controlled. The availability of Sema7a antibody is a big plus, as it allows to address the endogenous protein localization as well to show the signal absence in the sema7a mutant. The quantification of the arbor topology should be useful to people in the field who are looking at the lateral line as well as other axonal terminals. I think some results are overinterpreted though. The authors state: "Our findings demonstrate that Sema7A functions both as a juxtracrine and as a secreted cue to pattern neural circuitry during sensory organ development." However, they have not actually demonstrated which isoform functions in HCs (also see comments below).*

Comments: Thank you for making this point. To investigate the presence of both sema7a transcripts in the hair cells of the lateral-line neuromasts, we used the Tg(myo6b:actb1EGFP) transgenic fish to capture the labeled hair cells by fluorescence-activated cell sorting (FACS) and isolated total RNA. Using transcript specific DNA oligonucleotide primers, we have identified the presence of both sema7a transcript variants in the hair cell of the neuromast. Even though we have not developed transcript specific knockout animals, we speculate that the presence of both transcript variants in the hair cell implies that they function in distinct fashion. We have changed our interpretation in lines 32-34 to “Our findings propose that Sema7A likely functions both as a juxtracrine and as a secreted cue to pattern neural circuitry during sensory organ development.”

In future we will utilize the CRISPR/Cas9 technique to target the unique C-terminal domain of the GPI-anchored sema7a transcript variant. We believe that this will only perturb the formation of the full-length Sema7A protein and help us determine the role of the membrane-bound Sema7AGPI molecule as well as the Sema7Asec in sensory arborization and synaptic assembly.

> *In addition, they have to be careful in interpreting their topology analysis, as they cannot separate individual axons. Thus, such analysis can generate artifacts. They can perform additional experiments to address these issues or adjust their interpretations.*

Comments: Thank you for this insightful comment. In a previous eLife publication from our laboratory, we utilized the serial blockface scanning electron micrograph (SBFSEM) technique to characterize the connectome of the neuromast microcircuit where patterns of innervation of all the individual axons can be delineated in five-days-old larvae (reference 8). However, the collective behavior of all the sensory axons that build the innervation network remained enigmatic, especially in a living animal during development. In this paper we addressed how the sensory-axon collective behaves around the clustered hair cells and build the innervation network in living animals during diverse developmental stages. Our analyses have not only identified how the axons associates with the hair cell cluster as the organ matures, but also discovered distinct topological features in the arbor network that emerges during organ maturation, which may influence assembly of postsynaptic aggregates (lines

384-403, Figure 6G-I). We believe that our quantitative approach to capture collective axonal behaviors and their topological attributes during circuit formation have highlighted the importance of understanding network assembly during sensory organ development.

> ***Reviewer #3 (Public Review):***
>
> *Summary:*
>
> *This study demonstrates that the axon guidance molecule Sema7a patterns the innervation of hair cells in the neuromasts of the zebrafish lateral line, as revealed by quantifying gain- and loss-of function effects on the three-dimensional topology of sensory axon arbors over developmental time. Alternative splicing can produce either a diffusible or membrane-bound form of Sema7a, which is increasingly localized to the basolateral pole of hair cells as they develop (Figure 1). In sema7a mutant zebrafish, sensory axon arbors still grow to the neuromast, but they do not form the same arborization patterns as in controls, with many arbors overextending, curving less, and forming fewer loops even as they lengthen (Figure 2,3). These phenotypes only become significant later in development, indicating that Sema7a functions to pattern local microcircuitry, not the gross wiring pattern. Further, upon ectopic expression of the diffusible form of Sema7a, sensory axons grow towards the Sema7a source (Figure 4). The data also show changes in the synapses that form when mutant terminals contact hair cells, evidenced by significantly smaller pre- and post-synaptic punctae (Figure 5). Finally, by replotting single cell RNA-sequencing data (Figure 6), the authors show that several other potential cues are also produced by hair cells and might explain why the sema7a phenotype does not reflect a change in growth towards the neuromast. In summary, the data strongly indicate that Sema7a plays a role in shaping connectivity within the neuromast.*
>
> *Strengths:*
>
> *The main strength of this study is the sophisticated analysis that was used to demonstrate fine-level effects on connectivity. Rather than asking "did the axon reach its target?", the authors asked "how does the axon behave within the target?". This type of deep analysis is much more powerful than what is typical for the field and should be done more often. The breadth of analysis is also impressive, in that axon arborization patterns and synaptic connectivity were examined at 3 stages of development and in three-dimensions.*
>
> *Weaknesses:*
>
> *The main weakness is that the data do not cleanly distinguish between activities for the secreted and membrane-bound forms of Sema7a, which the authors speculate may influence axon growth and synapse formation respectively. The authors do not overstate the claims, but it would have been nice to see some additional experimentation along these lines, such as the effects of overexpressing the membrane-bound form,*

Comments: We have accepted this useful suggestion. In lines 333-346 and in Figure 5 we have demonstrated the impact of overexpressing the membrane-bound transcript variant on arborization pattern of the sensory axons.

> *Some analysis of the distance over which the "diffusible" form of Sema7a might act (many secreted ligands are not in fact all that diffusible), or*

Comments: We have reported this in lines 311-317 and in Figure 4F,G.

> *Some live-imaging of axons before they reach the target (predicted to be the same in control and mutants) and then within the target (predicted to be different).*

Comments: We have accepted this useful suggestion. We demonstrate the dynamics of the sensory arbors that are attracted to an ectopic Sema7Asec source in lines 325-332, Figure 4I,J; Figure 4—figure supplement 2A, and Videos 13-16.

> *Clearly, although the gain-of-function studies show that Sema7a can act at a distance, other cues are sufficient. Although the lack of a phenotype could be due to compensation, it is also possible that Sema7a does not actually act in a diffusible manner within its natural context. Overall, the data support the authors' carefully worded conclusions. While certain ideas are put forward as possibilities, the authors recognize that more work is needed. The main shortcoming is that the study does not actually distinguish between the effects of the two forms of Sema7a, which are predicted but not actually shown to be either diffusible or membrane linked (the membrane linkage can be cleaved). Although the study starts by presenting the splice forms, there is no description of when and where each splice form is transcribed.*

Comments: We have utilized the HCR™ RNA-FISH Technology to generate transcript specific probes. To generate transcript-specific HCR probes to distinctly detect the sema7aGPI (NM_001328508) and the sema7asec (NM_001114885) transcripts, Molecular Instruments could design only 11 probes against the sema7aGPI transcript and only one probe against the sema7asec transcript (personal correspondence with Mike Liu, PhD, Head of Operations and Product Development Lead Molecular Instruments, Inc.). The HCR probe against the sema7aGPI transcript showed a very faint signal. Unfortunately, the HCR probe against the sema7asec transcript failed to detect the presence of any transcript. For robust detection of transcripts, the protocol demands a minimum of 20 probes. We believe that the very low number of probes against our transcripts is the primary reason for the absence of a signal.

We therefore utilized fluorescence-activated cell sorting (FACS) to capture the labeled hair cells and isolated total RNA to perform RT-PCR using transcript specific DNA oligonucleotide primers. We identified the presence of both the secreted and the membrane-bound transcripts at four-days-old neuromasts (lines 80-84, Figure 1B-D).

> *Additionally, since the mutants are predicted to disrupt both forms, it is a bit difficult to disentangle the synaptic phenotype from the earlier changes in circuit topology - perhaps the change at the level of the synapse is secondary to the change in topology.*

Comments: Thank you for the insightful suggestion. We have analyzed the relationship between the sensory arbor network topology and the distribution of postsynaptic structures (lines 384-403, Figure 6G-I). We identified that the distribution of the postsynaptic aggregates is closely associated with the topological attributes of the sensory circuit. We further clarify the potential origin of disrupted synaptic assemblies in sema7a-/- mutants in lines 380-382 and lines 417-420.

> *Further, the authors do not provide any data supporting the idea that the membrane bound form of Sema7a acts only locally. Without these kinds of data, the authors are unable to attribute activities to either form.*

Comments: We have accepted this useful suggestion and have prepared the Figure 5 with the necessary details.

> *The main impact on the field will be the nature of the analysis. The field of axon guidance benefits from this kind of robust quantification of growing axon trajectories, versus their ability to actually reach a target. This study highlights the value of more careful analysis and as a result, makes the point that circuit assembly is not just a matter of painting out paths using chemoattractants and repellants, but is also about how axons respond to local cues. The study also points to the likely importance of alternative splice forms and to the complex functions that can be achieved using different forms of the same ligand.*
>
> ***Reviewer #4 (Public Review):***
>
> *Summary:*
>
> *The work by Dasgupta et al identifies Sema7a as a novel guidance molecule in hair cell sensory systems. The authors use the both genetic and imaging power of the zebrafish lateralline system for their research. Based on expression data and immunohistochemistry experiments, the authors demonstrate that Sema7a is present in lateral line hair cells. The authors then examine a sema7a mutant. In this mutant, Sema7a proteins levels are nearly eliminated. Importantly, the authors show that when Sema7a is absent, afferent terminals show aberrant projections and fewer contacts with hair cells. Lastly the authors show that ectopic expression of the secreted form of Sema7a is sufficient to recruit aberrant terminals to non-hair cell targets. The sema7a innervation defects are well quantified. Overall, the paper is extremely well written and easy to follow.*
>
> *Strengths:*
>
> *(1) The axon guidance phenotypes in sema7a mutants are novel, striking and thoroughly quantified.*
>
> *(2) By combining both loss of function sema7a mutants and ectopic expression of the secreted form of Sema7a the authors demonstrate the Sema7a is both necessary and sufficient to guide sensory axons*
>
> *Weaknesses:*
>
> *(1) Control. There should be an uninjected heatshock control to ensure that heatshock itself does not cause sensory afferents to form aberrant arbors. This control would help support the hypothesis that exogenously expressed Sema7a (via a heatshock driven promoter) is sufficient to attract afferent arbors.*

Comments: Thank you for the suggestion. We have added the uninjected heatshock control experiment in Figure 5 and described experimental details in the text, lines 343-345.

> *(2) Synapse labeling. The numbers obtained for postsynaptic labeling in controls do not match up with the published literature - they are quite low. Although there are clear differences in postsynaptic counts between sema7a mutants and controls, it is worrying that the numbers are so low in controls. In addition, the authors do not stain for complete synapses (pre- and post-synapses together). This staining is critical to understand how Sema7a impacts synapse formation.*

Comments: Thank you for raising this issue. We believe the low average numbers of the postsynaptic punctae in control neuromasts arise from lack of formation of postsynaptic aggregates beneath the immature hair cells, which are abundant in early stages of neuromast maturation. We have performed exhaustive analysis on the formation of pre- and postsynaptic structures and have identified how their distribution changes along neuromast development in control larvae. We have further analyzed how such distribution is perturbed in the sema7a-/- mutants. We do not think analyzing the complete synapse structure will add much to our understanding of how Sema7A influence synapse formation and maintenance.

> *(3) Hair cell counts. The authors need to provide quantification of hair cell counts per neuromast in mutant and control animals. If the counts are different, certain quantification may need to be normalized.*

Comments: We have added the raw data with the hair cell counts in both control and sema7a-/- mutants across developmental stages. The homozygous sema7a-/- mutants have slightly less hair cells and we have normalized all our topological analyses by the corresponding hair cell numbers for each neuromast in each experiment (lines 669-675).

> *(4) Developmental delay. It is possible that loss of Sema7a simply delays development. The latest stage examined was 4 dpf, an age that is not quite mature in control animals. The authors could look at a later age, such as 6 dpf to see if the phenotypes persist or recover.*

Comments: The homozygous sema7a-/- mutants are unviable and die at 6 dpf. We therefore restricted our analysis till 4 dpf. The association of the sensory arbors with the clustered hair cells gradually decreases as the neuromasts mature from 2 dpf to 4dpf in the sema7a-/- mutants (lines 174-176, Figure 2I). Moreover, in the sema7a-/- mutants the sensory axons throw long projections that keep getting farther away from the clustered hair cells as the neuromast matures from 2 dpf to 4 dpf (lines 166-168, Figure 2H; Figure 2—figure supplement 1K,L). These observations suggest that if the phenotypes in the sema7a-/- mutants were due to developmental delays, then we should have seen a recovery of disrupted arborization patterns over time. But instead, we observe a further deterioration of the arborization patterns and other architectural assemblies. These findings confirm that the observed phenotypes in the sema7a-/- mutants are not due to delayed development of the larvae, but a specific outcome for the loss of Sema7A protein.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Major concerns:*
>
> *Issue 1: One of the most interesting conclusions in this manuscript is the function of the GPIanchored vs. secreted form of Sema7a in axon structure and synapse formation. In lines 357360 of the discussion (for example) the authors state that they have shown that the GPIanchored form of Sema7a is responsible for contact-mediated synapse formation while the secreted form functions as a chemoattractant for axon arbor structure. "We have discovered dual modes of Sema7A function in vivo: the chemoattractive diffusible form is sufficient to guide the sensory arbors toward their target, whereas the membrane-attached form likely participates in sculpting accurate neural circuitry to facilitate contact-mediated formation and maintenance of synapses." However, the data do not support this conclusion. Specifically, no analysis is done showing unique expression of either isoform in hair cells and no functional analysis is done to conclusively determine which isoform is important for either phenotype.*

Comments: We have shown that both sema7a transcripts are expressed in the hair cells of four-day-old neuromasts (lines 78-84, Figure 1C,D). Ectopic expression of the sema7asec transcript variant robustly attracts the lateral-line sensory arbors toward itself, whereas ectopic expression of the sema7aGPI variant fails to impart sensory guidance from a distance, suggesting that the membrane-bound form likely participates in contact-mediated neural guidance. These experiments decisively show, for the first time in zebrafish, the dual modes of Sema7A function in vivo. However, we agree that the sema7aGPI transcript-specific knockout animal would be essential to conclusively prove that the membrane-attached form is primarily involved in forming accurate neural circuitry and contact-mediated formation and maintenance of synapses. Hence, we have very carefully stated in lines 427-428 that “the membrane-attached form likely participates in sculpting accurate neural circuitry to facilitate contact-mediated formation and maintenance of synapses”. We will follow up on this suggestion in our upcoming manuscript that will incorporate transcript-specific genetic ablations.

> *Though the authors present RT-PCR analysis of sema7a isoforms, it is not interpretable. The second reverse primer will also recognize the full-length transcript (from what I can gather) so it does not simply show the presence of the secreted form. Is there a unique 3'UTR for the short transcript that can be used? Additionally, for the GPI-anchored version can you use a forward primer that is not present in the short isoform? This would shed some light on the respective levels of both transcripts.*

Comments: The C-termini of the two transcript variants are distinct and we have designed distinct primers that will selectively bind to each transcript (lines 503-511). Since, we have not performed quantitative polymerase chain reaction (qPCR), relative levels of each transcript are hard to determine.

> *Alternatively, and perhaps of more use, in situ hybridization using unique probes for each isoform would allow you to determine which are actually present in hair cells.*

Comments: We have tried this approach and explained the point earlier (refer to lines 203212 of this response letter).

> *To decisively state that these isoforms have unique functions in axon terminal structure and synapse formation, other experiments are also essential. For example, RNA-mediated rescue analyses using both isoforms would tell you which can rescue the axonal structure and synapse size/number phenotypes. Overexpression of the GPI-anchored form, like the secreted form in Figure 4, would allow you to determine if only the secreted form can cause abnormal axon extension phenotypes. Expression of both forms in hair cells (using a myo6b promotor for example) would allow assessment of their role in presynapse formation.*

Comments: We have ectopically expressed the sema7aGPI transcript variant near the sensory arbor network and observed that Sema7A-GPI fails to impart sensory axon guidance from a distance.

Thank you for suggesting the rescue experiments. We are in the process of generating CRISPR/Cas9-mediated transcript-specific knockout animals. We are currently preparing another manuscript that incorporates the above-mentioned rescue experiments to dissect the role of each transcript in regulating arbor topology and synapse formation.

> *For the overexpression experiments, expression of mKate alone (with and without heat shock) is also a critical control to include.*

Comments: We have incorporated two control experiments: (1) larvae injected with hsp70:sema7asec-mKate2 plasmid that were not heat shocked and (2) Uninjected larvae that were heatshocked. We think these two controls are sufficient to demonstrate that the abnormal arborization patterns are not artifacts generated due to plasmid injection and heatshocking.

> *Issue 2: A second concern is the lack of data showing support cell and hair cell formation and function is unaffected. Analysis of support and hair cell number with loss of Sema7a as well as simple analyses of mechanotransduction (FM4-64) would help alleviate concerns that phenotypes are due to disrupted neuromast formation and basic hair cell function rather than a specific role for Sema7a in this process.*

Comments: We have measured the hair cell numbers in both control and sema7a-/- mutants across developmental stages. We have added this to our submitted raw data.

We have utilized the styryl fluorophore FM4-64 to test the mechanotransduction function of the hair cells in sema7a-/- mutants. We have detailed our finding in lines 137141 and in Figure 2—figure supplement 1C,D.

> *Expression analysis of Sema7a receptors would also help strengthen the argument for a specific effect on lateral line afferent axons.*

Comments: Thank you for this suggestion. Currently, we do not possess an RNA transcriptome dataset for the lateral line ganglion. This deficit limits a systematic screen for lateral-line sensory neuronal gene expressions either through antibody stains or via HCRmediated in situ techniques. In future we plan to develop an RNA transcriptome for the lateral-line ganglion and identify potential binding partners for Sema7A.

> *Issue 3: The manuscript could also be improved to include more detail in some areas and less in others. In general, each section has a fairly long lead up but lacks important experimental details that would help the reader interpret the data. For example:*
>
> *Figure 1: What is the label for the lateral line axons? Is it a specific transgenic? The legend states that 3 asterisks indicate p<0.0001. What about the other asterisk combinations?*

Comments: We have clarified these issues in lines 118-121 and in lines 906-907.

> *Figure 2: For the network analysis, are the traces for all axons that branch to innervate the neuromast?*

Comments: Yes, we have traced the entire arbor containing all the axons that branched from the lateral line nerve and extended toward the clustered hair cells. The three-dimensional traces depict a skeletonized representation of the arbor network.

> *Can the tracing method distinguish individual axons?*

Comments: No, our goal is to understand how the axon-collective behave around the clustered hair cells during development.

> *How do you know where an end is versus continued looping?*

Comments: We have categorically defined the topological attributes in lines 187-191 and in Figure 3A.

> *Also, are all neuromasts similarly affected or is there a divergence based on which organ you are imaging? What neuromast was imaged in this and other figures?*

Comments: Yes, all the neuromasts in the trunk and tail regions were affected similarly by the sema7a mutation. We did not observe any region-specific phenotypic outcome. We consistently imaged the trunk neuromasts, particularly the second, third, and fourth neuromasts.

> *Discussion: The short discussion failed to put these findings into context or to discuss how this unique topological arrangement of axon terminals impacts function.*

Comments: We have added a new segment, lines 432-448, in the discussion section which mentions the potential role of the topological features in arranging the distribution pattern of the postsynaptic densities and thereby potentially influencing the network's ability to gather sensory inputs through properly placed postsynaptic aggregates.

> *Can you speculate on how the looping structure may alter number of synaptic contacts per axon for instance? For this, it would be useful to know if normally the synapses form on loops versus bare terminals.*

Comments: Thank you for this insightful suggestion. We have performed detailed analysis, as mentioned in lines 384-397, to characterize the distribution of the postsynaptic densities between the two topological attributes.

> *Does this looping facilitate single axons contacting more hair cells of the same polarity? Would that be beneficial?*

Comments: Looping behaviors indeed facilitate the contact between the axons and the hair cells. As we have observed, the primary topological attribute that the sensory arbor network underneath the clustered hair cells adopts is a loop. The bare terminals are predominantly projected transverse to the clustered hair cells and lack contact with them. Whether a single axon, being part of a loop, preferentially contacts hair cells of same polarity is yet to be determined. We can address this question by mosaic labeling a single axon in the arbor network and determine its association with the hair cells. We intend to do these experiments in our upcoming manuscript.

> *Minor concerns:*
>
> *(1) For the stacked charts quantifying topological features, I found interpreting them challenging. Is it possible to put these into overlapping histograms or line graphs to better compare wild type to mutant directly?*

Comments: Thank you for your suggestion. We tried several ways to represent our data and found that the stacked charts optimally signify our analysis and depict the characteristic phenological differences between the control and the sema7a-/- mutants.

> *(2) There are numerous strong statements throughout not directly supported by the data, e.g. lines 110-113; 206-208; 357-360 and others. These should be tempered.*

Comments: For lines 110-113, we have updated this section with new experiments and the new segment is represented in lines 115-126.

For lines 206-208, we have updated the statement to "This result suggests that the stereotypical circuit topology observed in the mature organ may emerge through transition of individual arbors from forming bare terminals to forming closed loops encircling topological holes" in lines 225-227.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *The authors should be careful about making any assumptions which form of sema7a is active in NMs. Their RT-PCR demonstrates presence of both isoforms in a whole animal; however, whether they are similarly present in HCs is not investigated here.*

Comments: We have addressed this concern and have updated the manuscript with new experiments, detailed in lines 78-84.

> *Also, there is an issue of translation and trafficking to the membrane with subsequent secretion. An important experiment that would address this question is expressing two sema7a isoforms in mutant HCs and asking whether this can suppress the mutant phenotype.*

Comments: Thank you for suggesting the rescue experiments. We are in the process of generating CRISPR/Cas9-mediated transcript-specific knockout animals. We are currently preparing another manuscript that incorporates the above-mentioned rescue experiments to dissect the role of each transcript in regulating arbor topology and synapse formation.

> *Presumably, sema7a is trafficked to the membrane during HC maturation. This is consistent with the authors' observation that sema7a localization is changing as NM mature. However, actin-sema7a co-labeling does not actually show whether sema7a is on the membrane. Labeling HCs with a membrane marker (transgene) would be much more convincing. Alternatively, can the authors show sema7a localization actually correlates with the presence of sensory axon terminals? They already have immunos that label both. Thus, this should be pretty straightforward.*

Comments: Thank you for these suggestions. We have addressed these issues in lines 112114, and in lines 119-126.

> *Figure 2 should have a control panel, so the reduced sema7a staining can be compared to the control side-by-side.*

Comments: We have depicted Sema7A staining in control neuromasts in multiple images, including Figure 1E, Figure 1H, and in Figure 2—figure supplement 1B. We have kept the control panel in the supplementary figure due to space restrictions in Figure 2.

> *Arborization topology: While I appreciate the very careful characterization of the topology for wild-type and mutant NMs, I think it would be much more informative to mark individual axons and then analyze their topology. The main reason is that the authors cannot really distinguish whether some aspects of topology they describe are really due to the densely packed overlapping terminals of multiple axons or these are really characteristic, higher order organization of individual axons. Because of this, they cannot be certain what is really happening with sema7a mutant terminals. Related to the point above. While it is clear that the overall topology is abnormal in the mutant, the authors should be careful in concluding that sema7a regulates specific aspects of it. The overall structure is probably highly interconnected perturbing one parameter would likely affect all the others.*

Comments: Thank you for this comment. In a previous eLife publication from our laboratory, we utilized the serial blockface scanning electron micrograph (SBFSEM) technique to characterize the connectome of the neuromast microcircuit where patterns of innervation of all the individual axons can be delineated in five-days-old larvae (reference number 8). However, the collective behavior of all the sensory axons that build the innervation network remained enigmatic, especially in a living animal during development. In this paper we addressed how the sensory axon-collective behave around the clustered hair cells and build the innervation network in living animals during diverse developmental stages. Our analyses have not only identified how the axon-collective associates itself with the hair cell cluster as the organ matures, but also discovered distinct topological features in the arbor network that emerges during organ maturation, which may influence assembly of postsynaptic aggregates (lines 384-403, Figure 6G-I). We believe that our quantitative approach to capture collective axonal behaviors and their topological attributes during circuit formation have highlighted the importance of understanding network assembly during sensory organ development.

> *Experiments with the secreted sema7a isoform would be much more informative if they were compared/contrasted to the GPI anchored isoform.*

Comments: We added a new section, lines 338-351, and a new Figure 5 to address this issue.

> *The phenotype of ectopic projections in sema7a overexpression experiments is pretty dramatic, especially given the fact that these were performed in wild-type animals. Does this mean that the phenotype would be even more dramatic in sema7a mutants, as they have more bare axon terminals according to the authors' analysis. Have the authors attempted this type of experiments?*

Comments: That is an interesting suggestion. We have not tested that yet. Our guess is that in the sema7a-/- mutants, the abundant bare terminals will be far more sensitive to an ectopic source of Sema7A. But even in the sema7a-/- mutants, other chemotropic cues are still functional, which may impart certain restrictions on how many bare terminals are allowed to leave the neuromast region.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *(1) No raw data are shown, such that it is difficult to assess variability across animals or within animals, just the overall trends within the whole dataset. Raw data need to be shown for every measurement, at least in supplemental figures. It would also be useful to reliably show control next to mutant in the same plot, as it is a bit hard to compare across panels, which occurs in several figures.*

Comments: We have uploaded all the raw data related to each experiment.

> *(2) Given the focus on the two possible forms of Sema7a, the authors should use HCR or another form of reliable in situ hybridization to show the spatiotemporal pattern of expression of each isoform.*

Comments: We have utilized the HCR™ RNA-FISH Technology to generate transcript specific probes. To generate transcript-specific HCR probes to distinctly detect the sema7aGPI (NM_001328508) and the sema7asec (NM_001114885) transcripts, Molecular Instruments could design only 11 probes against the sema7aGPI transcript and only one probe against the sema7asec transcript (personal correspondence with Mike Liu, PhD, Head of Operations and Product Development Lead Molecular Instruments, Inc.). The HCR probe against the sema7aGPI transcript showed a very faint signal. Unfortunately, the HCR probe against the sema7asec transcript failed to detect the presence of any transcript. For robust detection of transcripts, the protocol demands a minimum of 20 probes. We believe that the very low number of probes against our transcripts is the primary reason for the lack of a signal.

> *(3) The authors should explain the criteria used to select the 22 embryos used to analyze the effects of expressing diffusible Sema7a.*

Comments: We have explained this in lines 291-292. We identified 22 mosaic sema7asecmKate2 integration events, in which a single mosaic ectopic integration had occurred near the network of sensory arbors, from a total of almost 100 integrations. We rejected events where the sema7asec-mKate2 integration occurred either farther away from the sensory arbor network or had happened in multiple neighboring cells.

> *(4) Although arbors were imaged in live embryos, time is never presented as a variable, so I cannot tell whether axon topology was changing as the images were collected. This needs to be clarified.*

Comments: We imaged the trunk neuromasts of both control and sema7a-/- mutant live zebrsfish larvae at 2, 3, and 4 dpf. We imaged the control and the sema7a-/- mutants of each developmental stage in parallel, within a span of two hours, and repeated these experiments multiple times to gather almost a hundred larvae from each genotype. Even though the sensory arbor network is dynamic, we believe imaging both the genotypes in parallel and within a span of two hours, and averaging almost a hundred larvae from each genotype minimize the temporal variability observed in the arbor architecture.

> *(5) Ideally, the authors should use CRISPR/cas-9 to create a mutation in the C-terminus that would prevent production of the GPI-anchored form and not of the diffusible form. I understand if this is too much work to do in a short time, and would be satisfied with another experiment that could distinguish roles for at least one isoform more clearly. For instance, it would be interesting to see an analysis of how far an axon can be from a source to detect diffusible Sema7a (live imaging would be ideal for this) and then to show that the effect is different when the membrane bound form is expressed.*

Comments: Thank you for this comment. We are currently working in generating transcript specific knockout animals.

We have added live timelapse video microscopy data in lines 330-337, Figure 4H-J, Figure 4—figure supplement 2, Video15,16.

We have added a new segment analyzing the membrane-bound transcript variant in lines 338-351.

> ***Reviewer #4 (Recommendations For The Authors):***
>
> *Feedback to authors*
>
> *Overall, this is a very important and novel study. Currently the manuscript does need revision.*
>
> *Major concerns:*
>
> *(1) Controls. For the ectoptic expression of Sema7a, injection of a construct expressing Sema7a under a heatshock promoter is used to drive ectopic expression. No heatshock (injected) animal are used as a control. In many systems heatshock can impact neuron morphology. And heatshock proteins are required for normal neurite and synapse formation. Please examine sensory axons in uninjected wildtype animals with heatshock.*

Comments: We have added this control experiment in a new segment, explained in detail in lines 348-350 and Figure 5.

> *(2) Synapse staining - regarding Figure 5 and related supplement*

> *Understanding whether guidance defects ultimately impact synapse formation is an important aspect of this paper. Therefore, is necessary to have accurate measurements of the number of complete synapses, and the overall numbers of pre- and postsynaptic components. Currently the data plotted in Figure 5 is extensive, but the way the data is laid out, the relevant comparisons are challenging to make. Perhaps include this quantification in the supplement, and move the data from the supplement to the main figure? The quantifications in the supplement are easier to follow and easier to compare between genotypes.*

Comments: We have performed exhaustive analysis on the formation of pre- and postsynaptic structures and have identified how their distribution changes along neuromast development in control larvae. We have further analyzed how such distribution is perturbed in the sema7a-/- mutants. We believe that showing only the average numbers will not reveal the changes in the distribution of the synaptic structures during development and across genotypes.

> *Looking at the data itself, there seems to be some discrepancies with the synaptic counts compared to published work. While the CTBP numbers seem in order, the Maguk numbers do not. In both mutant and control there are many hair cells without any Maguk puncta/aggregates-leading to 0.75-1 postsynapses per hair cell (Figure 5 supplement H-I). Typically, the numbers should be more comparable to what was obtained for CTBP, 3-4 puncta per cells (Figure 5 supplement B-C), especially by 3-4 dpf. 3-4 CTPB or Maguk puncta per cell is based on previously published immunostaining and EM work.*
>
> *The Maguk immunostaining, especially at early stages (2-3 dpf) is challenging. To compound a challenging immunostain, around 2019 Neuromab began to outsource the purification of their Maguk antibody. After this outsourcing our lab was no longer able to get reliable label with the Maguk antibody from Neuromab.*
>
> *Millipore sells the same monoclonal antibody and it works well: https://www.emdmillipore.com/US/en/product/Anti-pan-MAGUK-Antibody-clone-K2886,MM_NF-MABN72*
>
> *I would recommend this source.*

Comments: Thank you for suggesting the new MAGUK antibody. We have utilized this new MAGUK antibody from Millipore and added a new segment in lines 389-408. In future publication we will utilize this antibody to capture the postsynaptic densities in the sensory arbors.

The discrepancies in the postsynaptic punctae number in our control larvae may arise due to the reliability of the Neuromab MAGUK antibody. We have utilized this same antibody to stain the sema7a-/- mutants and have observed a significant decrease in MAGUK punctae number and area. On grounds of keeping parity between the control and the sema7a-/- mutants, we have decided to keep our experimental results in the manuscript.

In addition to a more accurate Maguk label, a combined pre- and post-synaptic label is essential to understand whether synapses pair properly in the sema7a mutants. This can be accomplished using subtype specific antibodies using goat anti-mouse IgG1/Maguk and goat anti-mouse IgG2a/CTBP secondaries.

Comments: Thank you for suggesting this. We are preparing another manuscript in which we will utilize this technique along with other suggestions to tease apart the role of distinct transcript variants in regulating neural guidance and synapse formation.

> *(3) Does sema7a lesion impact the number of hair cells per neuromast? If hair cell numbers are reduced several of the quantifications could be impacted.*

Comments: We have added the raw data with the hair cell counts in both control and sema7a-/- mutants across developmental stages. The homozygous sema7a-/- mutants have slightly less hair cells and we have normalized all our topological analyses by the corresponding hair cell numbers for each neuromast in each experiment (lines 669-675).

> *(4) Could innervation just be developmentally delayed in sema7a mutants? At 4 dpf the sensory system is just starting to come online and could still be in the process of refinement. Did you look at slightly older ages, after the sensory system is functional behaviorally, for example, 6 dpf? Do the cores phenotypes (synapse defects and excess arbors) persist at 6 dpf in the sema7a mutants?*

Comments: The homozygous sema7a-/- mutants are unviable and start to die at 6 dpf. We therefore restricted our analysis until 4 dpf. The association of the sensory arbors with the clustered hair cells gradually decreases as the neuromasts mature from 2 dpf to 4dpf in the sema7a-/- mutants (lines 174-176, Figure 2I). Moreover, in the sema7a-/- mutants the sensory axons throw long projections that keep getting farther away from the clustered hair cells as the neuromast matures from 2 dpf to 4 dpf (lines 166-168, Figure 2H; Figure 2—figure supplement 1K,L). These observations suggests that if the phenotypes in the sema7a-/- mutants were due to developmental delays, then we should have seen a recovery of disrupted arborization patterns over time. But instead, we observe a further deterioration of the arborization patterns and other architectural assemblies. These findings confirm that the observed phenotypes in the sema7a-/- mutants are not due to delayed development of the larvae, but a specific outcome for the loss of Sema7A protein.

> *Minor comments to address:*
>
> *Results*
>
> *Page 4 lines 89-91. For the readers, explain why you examined levels in Sema7a in rostral and caudal hair cells. Also, this sentence is, in general, a little bit misleading-initially reading that there is no difference in Sema7a at 1.5-4 dpf.*

Comments: In lines 44-48, we explain that the hair cells in the neuromast contain mechanoreceptive hair cells of opposing polarities that help them detect water currents from opposing directions. In lines 93-106, we tested whether the Sema7A level varies between the two polarities. We observed that the Sema7A level is similar between the two polarities of hair cells, but the average Sema7A intensity increases significantly over the developmental period of 2 dpf to 4 dpf in both rostrally and caudally polarized hair cells.

> *Page 10-11 Lines 263-270. What was the frequency of these 2 outcomes- out of the 22 cases with ectopic expression?*

Comments: We have explained this in lines 291-292. We identified 22 mosaic sema7asecmKate2 integration events, in which a single mosaic ectopic integration had occurred near the network of sensory arbors, from a total of almost 100 integrations. We rejected events where the sema7asec-mKate2 integration occurred either farther away from the sensory arbor network or had happened in multiple neighboring cells.

> *Discussion*

> *Page 14 Lines 359-360. There is not enough evidence provided in this work to suggest that the membrane attached form of Sema7a is playing a role. Both the secreted and membrane form are gone in the sema7a mutants. If the membrane attached form was specifically lesioned, and resulted in a phenotype, then there would be sufficient evidence. Currently there is strong evidence for a distinct role for the secreted form. Although the authors qualify the outlined statement with the word 'likely', stating this possibility in the discussion take-home is misleading.*

Comments: In future we will utilize the CRISPR/Cas9 technique to target the unique Cterminal domain of the GPI-anchored sema7a transcript variant. We believe that this will only perturb the formation of the full-length Sema7A protein and help us differentiate between the roles of the membrane-bound Sema7AGPI molecule and the secreted Sema7Asec in sensory arborization and synaptic assembly.

> *It might be interesting in either the intro or discussion to reference the role Sema3F in axon guidance in the mouse auditory epithelium. https://elifesciences.org/articles/07830*

Comments: We have added this reference in lines 61-64.

> *Figures*
>
> *Please indicate on one of your Figures where the mutation is (roughly) in the sema7a mutant (in addition to stating it in the results).*

Comments: We have added this information in Figure 2—figure supplement 1A.

> *Either state or indicate in a Figure where the epitope used to make the Sema7a antibody- to show that the antibody is predicted to recognize both isoforms.*

Comments: We have stated the details of the epitope in lines 528-529.

> *Figure 2-S1 what is the scale in panel A, is it different between mutant and wildtype?*

Comments: We have updated the images. New images are depicted in Figure 2—figure supplement 1A.

> *Methods*
>
> *What were the methods used to quantify synapse number and area?*

Comments: We have added a new section in lines 702-708 to explain the measurement techniques.

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